Q1

Find the relative error in P, if the physical quantity P is described by the relation $P = a^{\frac{1}{2}}\ b^2\ c^3 d^{-4}$. It is given that the relative errors in the measurement of a, b, c and d respectively, are 2%, 1%, 3% and 5%.

(1) 12%

(2) 8%

(3) 25%

(4) 32%

Q2

The centripetal acceleration of a particle varies inversely with the square of the radius r of the circular path .The KE of this particle varies directly as

(1) r

(2) r^2

(3) r^{-2}

(4) r^{-1}

Q3

A particle moves in $x - y$ plane according to rule $x\ =\ a \sin \omega t$ and $y\ =\ a\ \cos \omega t$. The particle follows

(1) An elliptical path

(2) A circular path

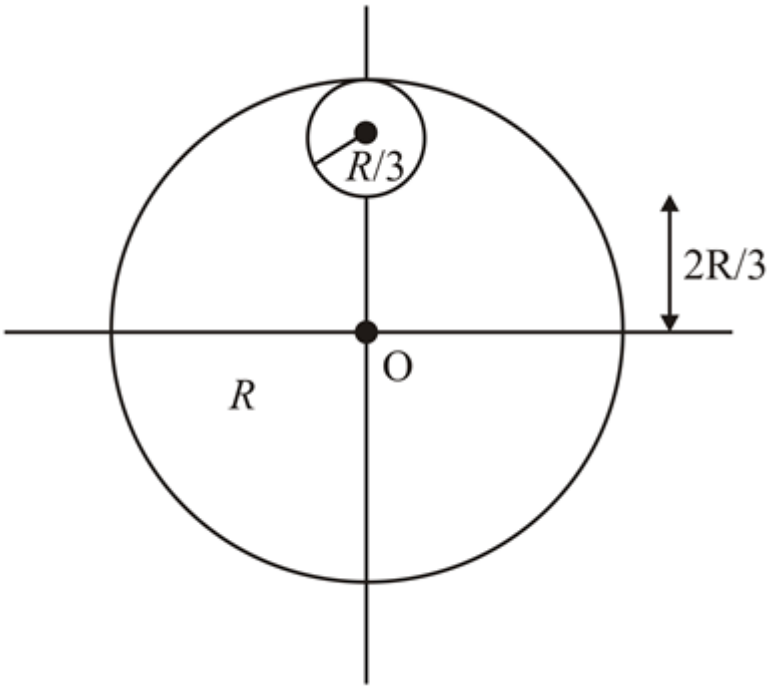
(3) A parabolic path

(4) A straight line path inclined equally to x and y-axes

Q4

From a circular disc of radius R and mass 9 M, a small disc of radius R/3 is removed from the disc. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing

through O is



(1) 4 MR^2

(2) $\frac{40}{9}MR^2$

(3) 10 MR^2

(4) $\frac{37}{9}MR^2$

Q5

The period of revolution of planet A around the Sun is 8 times that of B. The distance of A from the sun is n times greater than the distance of B from the sun. The value of n is

(1) 4

(2) 8

(3) 12

(4) 16

Q6

A drop of mercury of radius $2mm$ is split into 8 identical droplets. Find the increase in surface energy.(Surface tension of mercury is $0.465\ J/m^2$)

(1) $23.4\,\mu\ J$

(2) $18.5\,\mu J$

(3) $26.8\,\mu J$

(4) $16.8\,\mu J$

Q7

Assertion: Air quickly leaking out of a balloon becomes cooler.

Reason: The leaking air undergoes adiabatic expansion.

(1) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(2) If both Assertion and Reason are true but Reason is not explanation of the Assertion.

(3) If Assertion is true but the Reason is false.

(4) If Assertion is false but Reason is true.

Q8

Statement 1: For a charged particle moving from point P to point Q, the net work done by an electrostatic field on the particle is independent of the path connecting point P to point Q.

Statement 2: The net work done by a conservative force on an object moving along a closed loop is zero.

(1) Statement 1 is true, statement 2 is false.

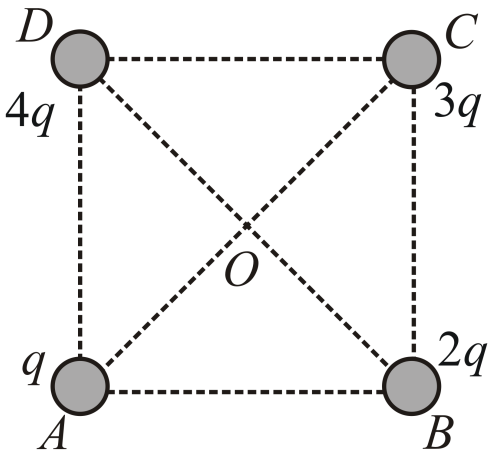
(2) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.

(3) Statement 1 is true, statement 2 is true; statement 2 is not the correct explanation for statement 1.

(4) Statement 1 is false, statement 2 is true.

Q9

Charges $q,\ 2q,\ 3q$ and $4q$ are placed at the corners $A,\ B,\ C$ and D of a square $ABCD$ as shown in the following figure. The direction of the electric field at the center of the square is along



(1) AB

(2) CB

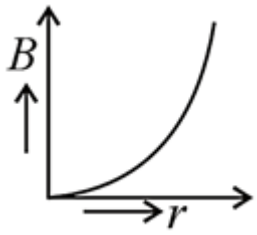
(3) BD

(4) AC

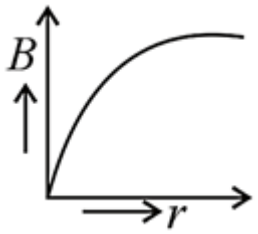
Q10

Which of the following graphs shows the variation of magnetic field B With distance for a long current carrying conductor.

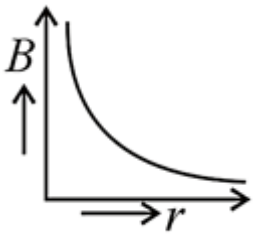
(1)



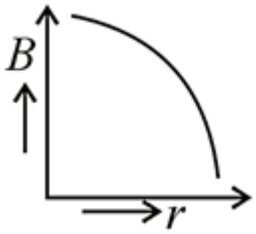
(2)



(3)



(4)



Q11

A SONAR (Sound Navigation And Ranging) uses ultrasonic waves to detect and locate objects under water. In a submarine equipped with a SONAR, the time delay between generation of a probe wave and the reception of its echo after reflection from an enemy submarine is found to be 77.0 s. What is the distance of the enemy submarine? Speed of sound in water $(v) = 1450 \text{ m s}^{-1}$.

(1) 55.825 km

(2) 55.21

(3) 25.55

(4) -55.32

Q12

The refracting angle of a prism is A and the refractive index of the material of the prism is $cot\left(\frac{A}{2}\right)$. The angle of minimum deviation of the prism is

(1) $\pi + 2A$

(2) $\pi - 2A$

(3) $\frac{\pi}{2} + A$

(4) $\frac{\pi}{2} - A$

Q13

A small telescope has an objective lens of focal length 144 cm and an eye piece of focal length 6 cm. Find the length of the telescope.

(1) 0.75 m

(2) 1.38 m

(3) 1.0 m

(4) 1.5 m

Q14

An $\alpha-$ particle and a proton are accelerated from rest by the same potential. The ratio of their de Broglie wavelengths would be:

[This question has been given as bonus]

(1) 2:1

(2) 1:2

(3) 1:3

(4) 3:1

Q15

Calculate the ground - state Q value of the induced fission reaction in the question

$n + {}^{235}_{92}U \rightarrow {}^{236}_{92}U^{*} \rightarrow {}^{90}_{40}Zr + {}^{134}_{52}Te + 3n$

If the neutron is thermal. A thermal neutron is in thermal equilibrium with its environment; it then has an average kinetic energy given by $\frac{3}{2}\ kT$.

(M_{Ur} -235.0439, M_{Zr}-98.9165 , M_{Te}-133.9115 , M_n -1.0087)

(1) 185 MeV

(2) 180 MeV

(3) 182 MeV

(4) 188 MeV

Q16

The ratio (R) of output resistance r_0, and the input resistance r_i in measurements of input and output characteristics of a transistor is typically in the range :

(1) R~$10^2 - 10^3$

(2) R~$1 - 10$

(3) R~$0.1 - 1.0$

(4) R~$0.1 - 0.01$

Q17

When $p - n$ junction diode is forward biased, then

(1) the depletion region is reduced and barrier height is increased

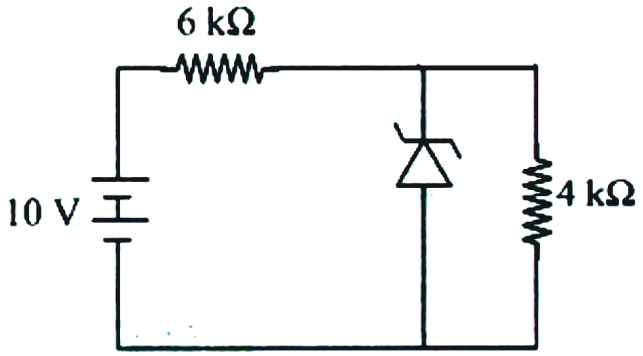
(2) the depletion region is widened and barrier height is reduced

(3) both the depletion region and barrier height are reduced

(4) both the depletion region and barrier height are increased

Q18

What will be the current flowing through the $6\text{k}\Omega$ resistor in the circuit shown, where the breakdown voltage of the zener is 6V?



(1) $\frac{2}{3}\text{mA}$

(2) $1\,\text{mA}$

(3) $10\,\text{mA}$

(4) $\frac{3}{2}\text{mA}$

Q19

Assertion: The receiver reconstructs the original message or data after its propagation through communication channel.

Reason: The reconstruction is necessary only in some cases.

(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) If both Assertion and Reason are true but Reason is not correct explanation of Assertion.

(3) If Assertion is true but Reason is false.

(4) If Assertion is false but Reason is true.

Q20

The velocity of sound in a gas, in which two waves of length $1.00\,\text{m}$ and $1.01\,\text{m}$ produce 10 beats in 3 seconds, is

(1) $336.67\ \mathrm{m\ s^{-1}}$

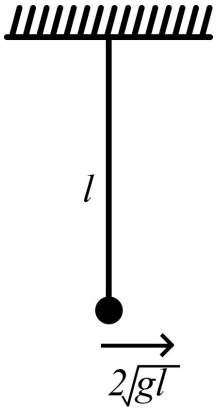
(2) $326.67\ \mathrm{m\ s^{-1}}$

(3) $346.67\ \mathrm{m\ s^{-1}}$

(4) $356.67\ \mathrm{m\ s^{-1}}$

Q21

One end of a string of length $\ell = \frac{14}{9}\mathrm{m}$ is fixed and a mass of 1 kg is tied to the other end. The ball is given a velocity $2\sqrt{g\ell}$ at the bottom most point as shown in figure. The string is cut when the ball becomes horizontal. Find the distance (in m) travelled till it stop for the 1st time $\left(\text{Take}\pi = \frac{22}{7}\right)$.

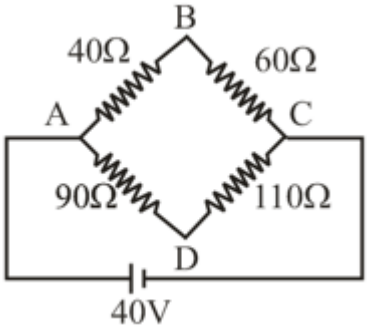


Q22

A resistance of 2 Ω is connected across one gap of a meter-bridge (the length of the wire is 100 cm) and an unknown resistance, greater than 2 Ω , is connected across the other gap. When these resistances are interchanged, the balance point shifts by 20 cm. Neglecting any end corrections, the unknown resistance is

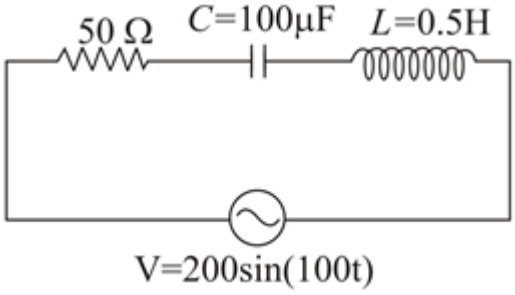
Q23

Four resistance 40 $\Omega, 60\ \Omega, 90\ \Omega$ 110 Ω and make the arms of a quadrilateral $ABCD$. Across AC is a battery of emf 40 V and internal resistance negligible. The potential difference across BD in V is _____



Q24

In the given $LCR\ AC$ circuit, the effective current (in ampere) flowing through the circuit is.

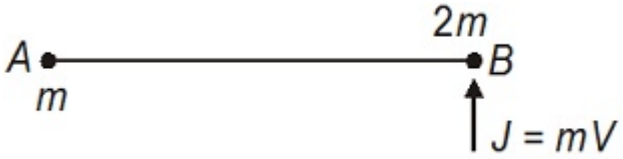


Q25

Two identical discs are positioned on a vertical axis. The bottom disc is rotating with angular velocity ω_0. The top disc is initially at rest. It is allowed to fall and sticks to the lower disc. Ratio of KE before and after collision.

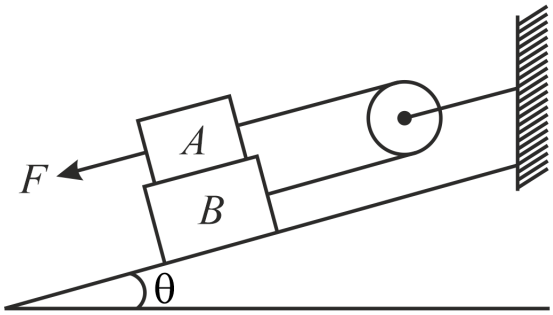
Q26

Two particles A and B of masses m and $2m$ are connected through a rod of length l. The system lies on a smooth horizontal surface. An impulse of magnitude mv to the particle B is imparted through impact horizontally but perpendicular to the rod. The angular velocity of the rod just after impact is $\frac{v}{nl}$. Find 2n.



Q27

Block A has a mass of 30 kg and block B a mass of 15 kg. The coefficients of friction between all surfaces of contact are $\mu_s = 0.15$ and $\mu_k = 0.10$. Knowing that $\theta = 30^\circ$ and that the magnitude of the force F applied to block A is 250 N, determine the acceleration of block A (in m s^{-2}). If acceleration is 'a', then the value of '5a' is .....



Q28

At constant volume, 4 mol of an ideal gas when heated from 300K to 500K changes its internal energy by 5000J. Let the molar heat capacity at constant volume is x. Then the value of 4x is .....

Q29

A charged particle is projected with velocity V_0 along positive $x-$ axis. The magnetic field B is directed along negative $z-$axis between $x = 0$ and $x = L$. The particle emerges out $(at\ x = L)$ at an angle of 60° with the direction of projection. The velocity with which the same particle is projected $(at\ x = 0)$ along positive $x-$ axis so that, when it emerges out $(at\ x = L)$ the angle made by it is 30° with the direction is pv_0 . Find the value of 2 p^2.

Q30

The value of inductance L for which the current is maximum in a series LCR AC circuit with $C = 10\ \mu$F and $\omega = 1000$ s^{-1} is $(20 \times n)$ mH, $n =$?

Q31

Which of the following statements is/are not true?

(1) Density of solid gets increased due to interstitial defects

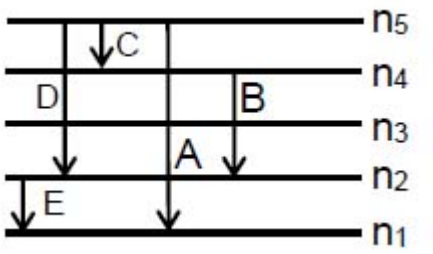
(2) Frenkel defects do not alter the density of the solid

(3) Non-stoichiometric defects modify the formula of the compound

(4) Non-stoichiometric defects do not alter the density of the solid

Q32

Spectral lines for H-atom were observed as shown below where n_1 to n_5 are successive shells. If 'E' belongs to visible region, then which of the following statements are correct for the transition?



(A) There cannot be any line in UV region

(B) B and D lines belongs to Infrared region.

(C) Line having shortest wavelength is A

(D) Line having least energy is C

(1) (A), (B) and (C)

(2) (A) and (B)

(3) (C) and (D)

(4) (A), (B), (C) and (D)

Q33

Which of the following is TRUE regarding hybridisation in PCl_5?

(1) All the bond angles in PCl_5 are equivalent.

(2) Axial and equatorial bonds have the same bond length.

(3) On orbital overlap between phosphorus and chlorine, five sp^3 d-p sigma covalent bonds are formed.

(4) Axial and equatorial bonds have equivalent bond energies.

Q34

Addition of sodium hydroxide solution to a weak acid (HA) results in a buffer of pH 6. If ionization constant of HA is 10^{-5}, the ratio of salt to acid concentration in the buffer solution will be:

(1) 10 : 1

(2) 4 : 5

(3) 5 : 4

(4) 1 : 10

Q35

Match List−I with List− II

	List−I (process)		List− II (catalyst)
(a)	Deacon's process	(i)	$ZSM-5$
(b)	Contact process	(ii)	$CuCl_2$
(c)	Cracking of hydrocarbons	(iii)	Ni
(d)	Hydrogenation of vegetable oils	(iv)	V_2O_5

Choose the most appropriate answer from the options given below-

(1) $a-ii, b-iv, c-i, d-iii$

(2) $a-i, b-ii, c-ii, d-iv$

(3) $a-iii, b-i, c-iv, d-ii$

(4) $a-iv, b-ii, c-i, d-iii$

Q36

Column I	Column II
A. He	i. Highest electron gain enthalpy
B. Cl	ii. Most electropositive element
C. Ca	iii. Strongest reducing agent
D. Li	iv. Highest ionisation energy

The correct match of the contents in column I with those in column II is

(1) A-(iii), B-(i), C-(ii), D-(iv)

(2) A-(iv), B-(iii), C-(ii), D-(i)

(3) A-(i), B-(ii), C-(iii), D-(iv)

(4) A-(iv), B-(i), C-(ii), D-(iii)

Q37

The froth-flotation process for concentration of sulphide ores is based upon

(1) The difference in magnetic properties of **gangue particles** and **ore particles**.

(2) The difference in specific gravity of **gangue particles** and **ore particles**.

(3) Preferrential wetting of **gangue particles** by oil

(4) Preferrential wetting of **ore particles**

Q38

Which is an electron deficient hydride among the following?

(1) LiH

(2) CaH_2

(3) AlH_3

(4) SiH_4

Q39

The main oxides formed on combustion of Li, Na and K in air are respectively

(1) Li_2O, Na_2O_2 and KO_2

(2) Li_2O, Na_2O and KO_2

(3) LiO_2, Na_2O_2 and K_2O

(4) Li_2O_2, Na_2O_2 and KO_2

Q40

Which oxo−acid of sulphur contains the S− S bond in its structure?

(1) Disulphurous acid

(2) Disulphuric acid

(3) Perdisulphuric acid

(4) Hyposulphurous acid

Q41

Among the following complexes the one which shows Zero crystal field stabilizations energy (CFSE)

(1) $\left[Nn(H_2O)_6\right]^{3-}$

(2) $\left[Fe(H_2O)_6\right]^{3+}$

(3) $\left[Co(H_2O)_6\right]^{2+}$

(4) $\left[Co(H_2O)_6\right]^{3+}$

Q42

The processes of calcination and roasting in metallurgical industries, respectively, can lead to :

(1) Global warming and photochemical smog

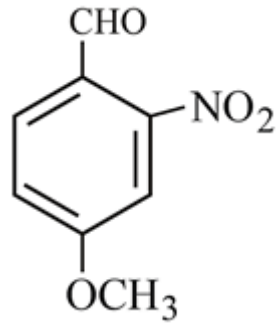
(2) Global warming and acid rain

(3) Photochemical smog and ozone layer depletion

(4) Photochemical smog and global warming

Q43

What is the correct IUPAC name of



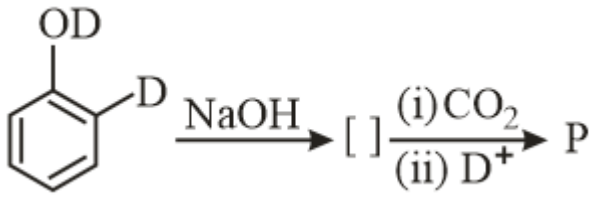
(1) 4-methoxy-2-nitrobenzaldehyde

(2) 4-formyl-3-nitro anisole

(3) 4-methoxy-6-nitrobenzaldehyde

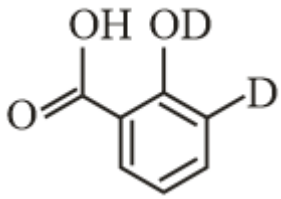
(4) 2-formyl-5-methoxy nitrobenzene

Q44

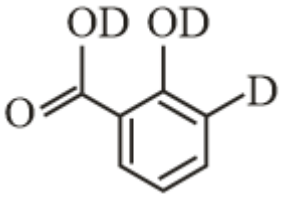


Here, P is;

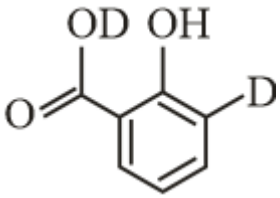
(1)



(2)



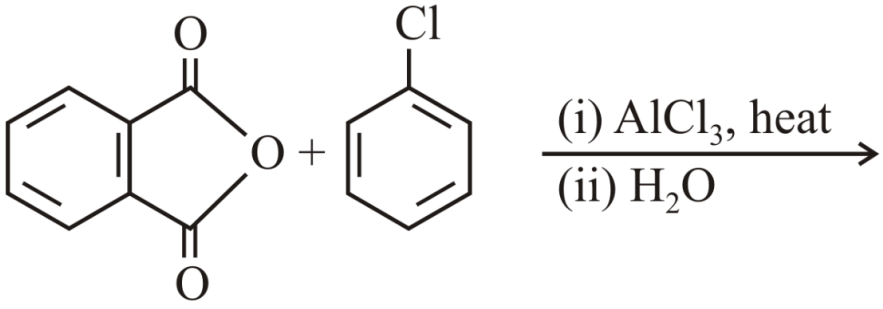
(3)



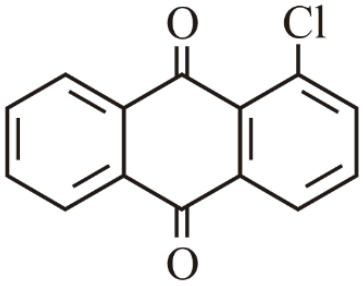
(4) Reaction not possible.

Q45

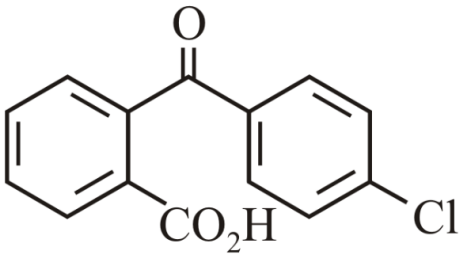
The major product of the following reaction is:



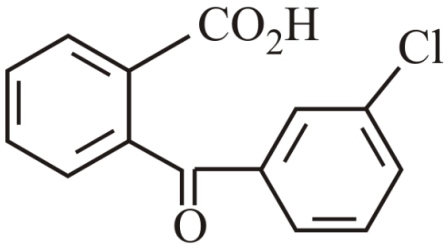
(1)



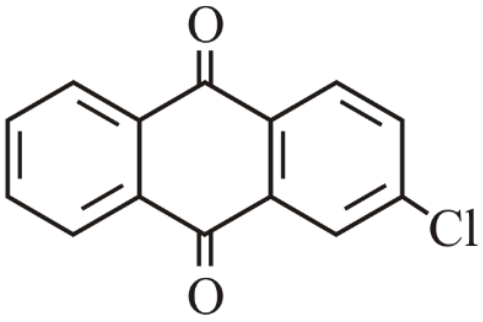
(2)



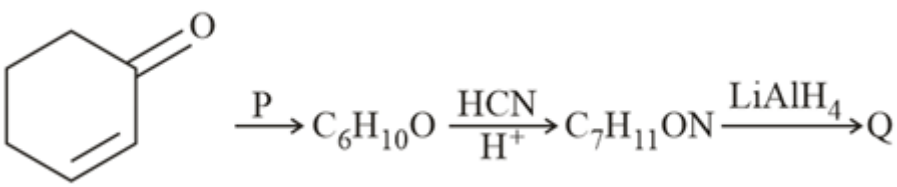
(3)



(4)

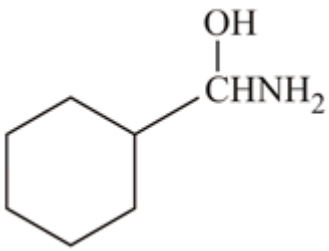


Q46



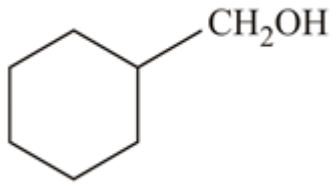
P and Q are -

(1) P is $NaBH_4$



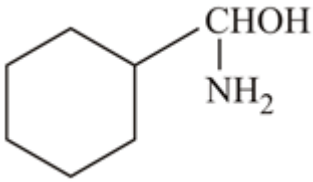
Q is

(2) P is $Ni-H_2$, in ethanol,



Q is

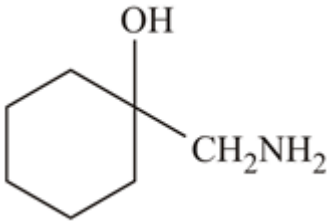
(3) P is $LiAlH_4$,



Q is

(4) P is $Pd-H_2$, in quinoline at room temperature

Q is



Q47

Given below are two statements :

Statement I : Aniline is less basic than acetamide.

Statement II : In aniline, the lone pair of electrons on nitrogen atom is delocalised over benzene ring due to resonance and hence less available to a proton.

Choose the most appropriate option ;

(1) Statement I is true but statement II is false.

(2) Statement I is false but statement II is true.

(3) Both statement I and statement II are true.

(4) Both statement I and statement II are false.

Q48

Consider the Assertion and Reason given below.

Assertion (A) : Ethene polymerized in the presence of Ziegler Natta Catalyst at high temperature and pressure is used to make buckets and dustbins.

Reason (R) : High density polymers are closely packed and are chemically inert. Choose the correct answer from the following:

(1) (A) is correct but (R) is wrong.

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(4) (A) and (R) both are wrong.

Q49

Which one of the following statements about the structure of double-helical DNA is incorrect ?

(1) Within the double helix there are 10 base pairs per turn of the helix.

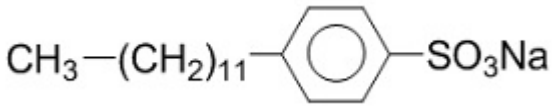
(2) The proces that stabilize the DNA double helix are hydrogen bonds between compementary bases.

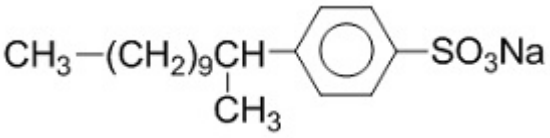
(3) Separation of the two strands of the double helix requires untwisting of the helix.

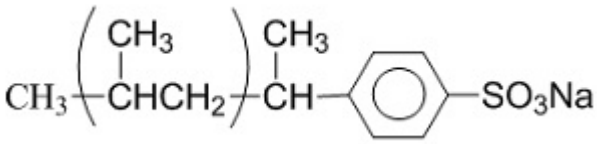
(4) The molar amount of adenine plus thymine equals the molar amount of guanine plus cytosine.

Q50

Which of the following is an example of a non-biodegradable detergent?

(1) 

(2) 

(3) 

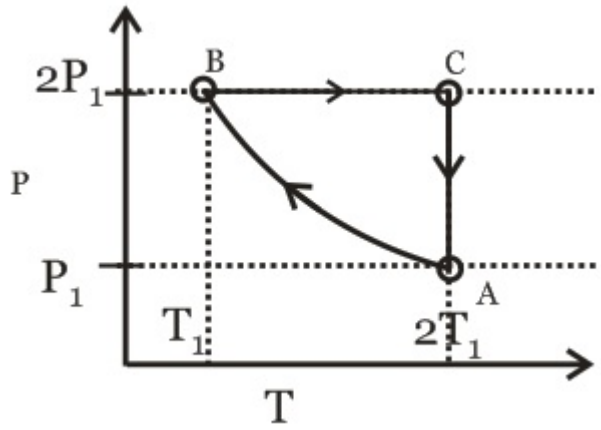
(4) $CH_3(CH_2)_{10}CH_2OSO_3Na$

Q51

Rate of diffusion of H_2 Gas is $3\sqrt{3}$ Times that of a hydrocarbon gas (C_nH_{2n-2}) Under identical conditions. Calculate value of n.

Q52

Two moles of an ideal gas (monatomic) are taken through cycle $ABCA$. During process $AB,\ PT = \text{constant}\ \ T1 = 300\ K$. The work done in the process may be $(150\ R)x$, the value of x is:



Q53

For the reaction

$$A + B \rightleftharpoons 2C$$

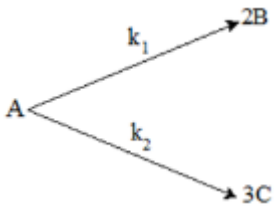
the value of equilibrium constant is 100 at 298 K. If the initial concentration of all the three species is 1 M each, then the equilibrium concentration of C is $x \times 10^{-1}$ M. The value of x is _____. (Nearest integer)

Q54

A KCl solution of conductivity $0.14\ S\ m^{-1}$ shows a resistance of 4.19Ω in a conductivity cell. If the same cell is filled with an HCl solution, the resistance drops to 1.03Ω. The conductivity of the HCl solution is ____ $\times 10^{-2}\ S\ m^{-1}$. (Round off to the Nearest Integer).

Q55

Consider the following parallel reactions being given by A ($t_{1/2} = 1.386 \times 10^2$ hours), each path being 1st order.



If distribution of B in the product mixture is 50%, calculate partial half-life of A for the conversion into B.

Q56

In neutral or faintly alkaline solution, 8 moles of permanganate anion quantitatively oxidize thiosulphate anions to produce X moles of a sulphur containing product. The magnitude of X is

Q57

How many of the following are optically inactive?

i. trans $-\left[\mathrm{Co}\left(\mathrm{en}\right)_2(\mathrm{NH_3})\mathrm{Cl}\right]^{2+}$

ii. cis $-\left[\mathrm{Co}\left(\mathrm{en}\right)_2\mathrm{Br_2}\right]^{+}$

iii. $\left[\mathrm{Co}\left(\mathrm{NH_3}\right)_3\mathrm{Cl_3}\right]$

iv. trans $-\left[\mathrm{Co}\left(\mathrm{NH_3}\right)_4\mathrm{Cl_2}\right]^{+}$

v. trans $-\left[\mathrm{CoCl_2}\left(\mathrm{C_2O_4}\right)_2\right]^{3-}$

Q58

The percentage of anhydrous salt is 56.25 by weight in the crystalline solid $\mathrm{MSO_4.nH_2O}$ of molar mass $288\ \mathrm{g\ mol^{-1}}$. The value of 'n' is:

Q59

How many ethers will be formed when a mixture of $\mathrm{C_2H_5OH}$ and methyl alcohol are treated with concentrated $\mathrm{H_2SO_4}$?

Q60

$15\ \mathrm{mL}$ of aqueous solution of $\mathrm{Fe^{2+}}$ in acidic medium completely reacted with $20\ \mathrm{mL}$ of $0.03\ \mathrm{M}$ aqueous $\mathrm{Cr_2O_7^{2-}}$. The molarity of the $\mathrm{Fe^{2+}}$ solution is _____ $\times 10^{-2}\mathrm{M}$ (Round off to the Nearest Integer).

Q61

If α and β are the roots of the equation $x^2-2x+3=0$, then the sum of roots of the equation having roots as $\alpha^3-3\alpha^2+5\alpha-2$ and $\beta^3-\beta^2+\beta+5$ is

(1) 1

(2) 3

(3) 5

(4) 7

Q62

Let $z_1,\ z_2$ and z_3 are the points on the argand plane which lie on the circle with equation $|z-z_0|=\frac{4}{\sqrt{3}}$ (where z_0 is the centre of the circle). If $z_1=0,\ z_2=-4$ and $z_3=4+3z_0$, then $|\arg(z_0)|$ is equal to (where $\arg Z\in(-\pi,\pi]$)

(1) $\frac{\pi}{6}$

(2) $\frac{5\pi}{12}$

(3) $\frac{5\pi}{6}$

(4) $\frac{2\pi}{3}$

Q63

Let P_n be a square matrix of order 3 such that $P_n=\left[a_{ij}\right]$, where $a_{ij}=\frac{3i+j}{4^{2n}}$ for $1\le i\le 3, 1\le j\le 3$. Then the value of $\lim\limits_{n\to\infty}T_r\left(4P_1+4^2P_2.....4^nP_n\right)$ is

(where $T_r(A)$ denotes trace of matrix A i.e sum of principal diagonal elements of A)

(1) 7

(2) 8

(3) $\frac{25}{3}$

(4) 9

Q64

The sum to the infinite terms of the series $\frac{5}{3^2\cdot7^2}+\frac{9}{7^2\cdot11^2}+\frac{13}{11^2\cdot15^2}+....$ is

(1) $\frac{1}{8}$

(2) $\frac{1}{36}$

(3) $\frac{1}{54}$

(4) $\frac{1}{72}$

Q65

The value of $\lim\limits_{x\to 0}\frac{x^{6000}-(\sin x)^{6000}}{x^2(\sin x)^{6000}}$ is

(1) 1000

(2) 100

(3) 1100

(4) 1010

Q66

If the function $f(x)= x^3 - 3ax$ has a local minimum at $x = \lambda$ $(\lambda \geq 4)$ and $a \in [10,18]$, then the sum of all the possible integral values of a is

(1) 50

(2) 112

(3) 51

(4) 16

Q67

Let $f : (0, 1) \to R$ be a function defined by $f(x) = 10x^{\sin x+\cos x}$. If $I = \int_0^1 f(x)dx$, then find the value of $[I]$.

[Note: Where $[k]$ denotes the greatest integer function less than or equal to k.]

(1) 3

(2) 5

(3) 7

(4) 4

Q68

Consider a function $f(x)$ which satisfies $f'(x) = \tan^2 x + \int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} f(x)dx$. Also $f\left(\frac{\pi}{4}\right) = \frac{-\pi}{4}$.

If the value of $\frac{8+\pi^2}{\pi}\int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} f(x)dx = m$, then find the value of m^2.

(1) 4

(2) 8

(3) 32

(4) 16

Q69

The curve having differential equation, $x\cos y\frac{dy}{dx}+\sin y=x$ and passing through the origin, also passes through

(1) $\left(2,\frac{\pi}{2}\right)$

(2) $\left(-2,\frac{\pi}{2}\right)$

(3) $\left(4,\frac{3\pi}{2}\right)$

(4) $\left(-8,\frac{3\pi}{2}\right)$

Q70

The normal to the parabola $y^2=4x$ at $P(9,6)$ meets the parabola again at Q. If the tangent at Q meets the directrix at R, then the slope of another tangent drawn from point R to this parabola is

(1) 11

(2) $\frac{11}{3}$

(3) $\frac{3}{11}$

(4) 3

Q71

Let PQ be the common chord of the circles $S_1: x^2+y^2+2x+3y+1=0$ and $S_2: x^2+y^2+4x+3y+2=0$, then the perimeter (in units) of the triangle $\mathrm{C_1PQ}$ is equal to $\left(\text{where, } C_1=\left(-1,\frac{-3}{2}\right)\right)$

(1) $\frac{9}{2}$

(2) $2\sqrt{2}+3$

(3) $3\sqrt{2}+2$

(4) $\frac{3}{2}+2\sqrt{2}$

Q72

Let $P_1: \ 3y+z+1=0$ and $P_2: \ 2x-y+3z-7=0$ and the equation of line AB is $\frac{x-1}{2}=\frac{y-3}{-1}=\frac{z-4}{3}$ in 3D space. Shortest distance between the line of intersection of planes P_1 and P_2 and the line AB is equal to

(1) $\frac{7}{\sqrt{10}}$ units

(2) $7\sqrt{\frac{2}{5}}$ units

(3) $\frac{6}{\sqrt{10}}$ units

(4) $2\sqrt{\frac{2}{5}}$ units

Q73

In ΔPQR, the equation of the internal angle bisector of angle Q is $y=x$ and the equation of side PR is $3x-y=2$. If coordinates of P are $(3,2)$ and $2PQ=RQ$, then the coordinates of Q are

[This question has been given as bonus]

(1) $(3,3)$

(2) $(7,7)$

(3) $(-2,-2)$

(4) $(5,5)$

Q74

A ray of light through the point $A(1,2,3)$ strikes the plane $x+y+z=12$ at a point B and on reflection passes through the point $C(3,5,9)$. If the equation of a plane containing the incident ray and the reflected ray is $P=0$ and the distance of $P=0$ from $(0,0,0)$ is λ units, then the value of $13\lambda^2$ is equal to

(1) 1

(2) 2

(3) 4

(4) 6

Q75

Let $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - \hat{k}$. If $\vec{r} \times \vec{a} = \vec{b}$ and $\vec{r} \cdot \vec{c} = 3$, then the value of $\left|\vec{r}\right|$ is equal to

(1) $\sqrt{155}$

(2) $\sqrt{17}$

(3) $2\sqrt{17}$

(4) 3

Q76

Let Z be the set of integers, if $A = \left\{x \in Z : \ |x-3|^{\left(x^2-5x+6\right)} = 1\right\}$ and $B = \{x \in Z : \ 10 < 3x + 1 < 22\}$, then the number of subsets of the set $A \times B$ is

(1) 2^6

(2) 2^8

(3) 2^{15}

(4) 2^9

Q77

The mean and variance of 10 observations are found to be 10 and 5 respectively. On rechecking it is found that an observation 5 is incorrect. If the incorrect observation is replaced by 15, then the correct variance is

(1) 7

(2) 8

(3) 9

(4) 4

Q78

A tower T_1 of height 60 m is located exactly opposite to a tower T_2 of height 80 m on a straight road. From the top of T_1, if the angle of depression of the foot of T_2 is twice the angle of elevation of the top of T_2, then the width (in m) of the road between the feet of the towers T_1 and T_2 is

(1) 20 $\sqrt{2}$

(2) 10 $\sqrt{2}$

(3) 10 $\sqrt{3}$

(4) 20 $\sqrt{3}$

Q79

Which of the following statements is not a tautology?

(1) $(p \wedge q \wedge r) \Rightarrow (\sim p \vee \sim q \vee r)$

(2) $(p \wedge q \wedge r) \Rightarrow ((\sim p \wedge \sim q) \vee r)$

(3) $(p \wedge \sim q \wedge r) \Rightarrow (\sim p \vee q \vee r)$

(4) $(p \wedge q \wedge \sim r) \Rightarrow r$

Q80

If $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \ldots + C_n x^n$, $\sum_{r=0}^{n} \left((r+1)^2\right) C_r = 2^{n-2} f(n)$ and if the roots of the equation $f(x) = 0$ are α and β, then the value of $\alpha^2 + \beta^2$ is equal to (where C_r denotes ${}^nC_r$)

(1) 13

(2) 10

(3) 17

(4) 20

Q81

Let $\vec{a}$ be a unit vector coplanar with $\hat{i} - \hat{j} + 2\hat{k}$ and $2\hat{i} - \hat{j} + \hat{k}$ such that $\vec{a}$ is perpendicular to $\hat{i} - 2\hat{j} + \hat{k}$. If the projection of $\vec{a}$ along $\hat{i} - \hat{j} + 2\hat{k}$ is λ, then the value of $\frac{1}{\lambda^2}$ is equal to

Q82

If for $x \geq 0, y = y(x)$ is the solution of the differential equation,

$(x+y)dy = \Big((x+1)^2 + y - 3\Big)dx, y(2) = 0$ then $y(3)$ is equal to _____

[This question has been given as bonus]

Q83

If x, y are positive real numbers and $3x + 4y = 5$, then the largest possible value of $16x^2y^3$ is

Q84

The maximum value of x that satisfies the equation $\sin^{-1}\left(\frac{2\sqrt{15}}{|x|}\right) = \cos^{-1}\left(\frac{14}{|x|}\right)$ is

Q85

The area (in sq. units) bounded by $[|x|]+[|y|] = 2$ in the first and third quadrant is (where, [.] is the greatest integer function),

Q86

Find the sum of all integral values of a for which all the roots of the equation $x^4 - 4x^3 - 8x^2 + a = 0$ are real.

Q87

Let Z be a complex number satisfying the relation $Z^3 + \frac{4\left(\overline{Z}\right)^2}{|Z|} = 0$. If the least possible argument of Z is $-k\pi$, then 5k is equal to (here, arg $Z \in (-\pi, \pi]$)

Q88

If the number of solutions of the equation $x + y + z = 20$, where $1 \leq x < y < z$ and $x, y, z \in I$ is k, then k is equal to

Q89

A is square matrix of order n.

$\ell =$ maximum number of distinct entries if A is a triangular matrix

$m =$ maximum number of distinct entries if A is a diagonal matrix

$p =$ minimum number of zeroes if A is a triangular matrix

If $\ell + 5 = \mathrm{p} + 2\mathrm{m}$, find the order of the matrix.

Q90

$S = \{1, 2, 3\}, f : S \to S$ satisfies the property : $\forall x \in S, f(f(x)) = f(x)$. How many different functions are there for $f(x)$?

Answer Key

Q1 (4)	**Q2** (4)	**Q3** (2)	**Q4** (1)
Q5 (1)	**Q6** (1)	**Q7** (1)	**Q8** (3)
Q9 (2)	**Q10** (3)	**Q11** (1)	**Q12** (2)
Q13 (4)	**Q14** (1)	**Q15** (1)	**Q16** (1)
Q17 (3)	**Q18** (1)	**Q19** (3)	**Q20** (1)
Q21 (4)	**Q22** (3)	**Q23** (2)	**Q24** (2)
Q25 (2)	**Q26** (4)	**Q27** (26)	**Q28** (25)
Q29 (6)	**Q30** (5)	**Q31** (4)	**Q32** (4)
Q33 (3)	**Q34** (1)	**Q35** (1)	**Q36** (4)
Q37 (4)	**Q38** (3)	**Q39** (1)	**Q40** (4)
Q41 (2)	**Q42** (2)	**Q43** (1)	**Q44** (3)
Q45 (2)	**Q46** (4)	**Q47** (2)	**Q48** (3)
Q49 (4)	**Q50** (3)	**Q51** (4)	**Q52** (8)
Q53 (25)	**Q54** (57)	**Q55** (231)	**Q56** (6)
Q57 (4)	**Q58** (7)	**Q59** (3)	**Q60** (24)
Q61 (2)	**Q62** (3)	**Q63** (2)	**Q64** (4)

Q65 (1)	**Q66** (3)	**Q67** (4)	**Q68** (4)
Q69 (1)	**Q70** (2)	**Q71** (2)	**Q72** (2)
Q73 (2)	**Q74** (2)	**Q75** (1)	**Q76** (1)
Q77 (4)	**Q78** (4)	**Q79** (4)	**Q80** (3)
Q81 (12)	**Q82** (3)	**Q83** (3)	**Q84** (16)
Q85 (6)	**Q86** (6)	**Q87** (3)	**Q88** (24)
Q89 (4)	**Q90** (10)		

Q1

$P = a^{\frac{1}{2}}\ b^2c^3d^{-4}$

$\left(\frac{\Delta p}{p}\right)_{max} = \frac{1}{2}\ \frac{\Delta a}{a} + 2\frac{\Delta b}{b} + 3\frac{\Delta c}{c} + 4\frac{\Delta d}{d}$

$100 \times \frac{\Delta P}{P} = \left(\frac{1}{2}\ \frac{\Delta a}{a} + 2\frac{\Delta b}{b} + 3\frac{\Delta c}{c} + 4\frac{\Delta d}{d}\right) \times 100$

$\%\ error\ in\ P = \left(\frac{1}{2} \times 2 + 2 \times 1 + 3\ \times 3 + 4\ \times 5\right)$

$= 32\%$

Q2

Centripetal acceleration

$\frac{v^2}{r} \propto r^{-2}$

Or $\frac{v^2}{r} = \frac{k}{r^2}$

Or $v^2 = \frac{k}{r}$

$\therefore\ \ KE = \frac{1}{2}mv^2$

$= \frac{1}{2}\ \frac{mk}{r}$

Or $KE \propto r^{-1}$

Q3

We have, $x = a\sin\omega t$

$\frac{x}{a} = \sin\omega t\ .....(i)$

and

$y = a\cos\omega t$

$\Rightarrow \frac{y}{a} = \cos\omega t \ldots\ldots (ii)$

Squaring and adding, we get

$\frac{x^2}{a^2} + \frac{y^2}{a^2} = 1 \qquad \left(\because \ \cos^2\omega t + \sin^2\omega t = 1\right)$

or $x^2 + y^2 = a^2$

This is the equation of a circle.

Hence, particle follows a circular path.

Q4

The moment of inertia of complete disc about a perpendicular axis passing through centre O is,

$I_1 = \frac{1}{2}(9M)\,R^2$

$= \frac{9}{2}MR^2$

The mass of cut out disc of radius $\frac{R}{3}$ is,

$m = \frac{9M}{\pi R^2}\pi\left(\frac{R}{3}\right)^2$

$= M$

Now, using the theorem of the parallel axis, the moment of inertia of cut out disc about the perpendicular axis passing through centre O is,

$I_2 = \frac{1}{2}M\left(\frac{R}{3}\right)^2 + M\left(\frac{2R}{3}\right)^2$

$= \frac{1}{2}MR^2$

The moment of inertia of residue disc is,

$I = I_1 - I_2$

$= \frac{9}{2}MR^2 - \frac{1}{2}MR^2$

Q5

The period of a satellite is the time it takes it to make one full orbit around an object. The period of a satellite are only dependent upon the radius of orbit and the mass of the central body that the satellite is orbiting.

As per Kepler's third law,

$T^2 \propto r^3 \Rightarrow \frac{r_{\mathrm{A}}}{r_{\mathrm{B}}}=\left(\frac{T_{\mathrm{A}}}{T_{\mathrm{B}}}\right)^{2/3}$

Given,

$T_A = 8T_B$

$\therefore \frac{r_A}{r_B}=\left(\frac{8T_B}{T_B}\right)^{2/3}=4=n$

Q6

Increase in surface energy

$=4\pi R^2 T\left(n^{1/3}-1\right)=4\pi\left(2\times10^{-3}\right)^2\left(0.465\right)\ \left(8^{1/3}-1\right)=23.4\times10^{-6}\,J=23.4\,\mu\ J$

Q7

When a system undergoes a change under the condition that there is no exchange of heat between the system and surroundings, then the process is called an adiabatic process. The leaking air of balloon undergoes adiabatic expansion. Air cools down due to adiabatic expansion as air has to do work against the external pressure at the cost of its internal energy. Due to work done against external pressure, the internal energy of air reduces. Thus, it becomes cooler.

Q8

When an object is under a conservative force moving from one position to another, then the value of the net work done depends only on the initial and final position of the object, i.e., the work done by a conservative force is path independent.

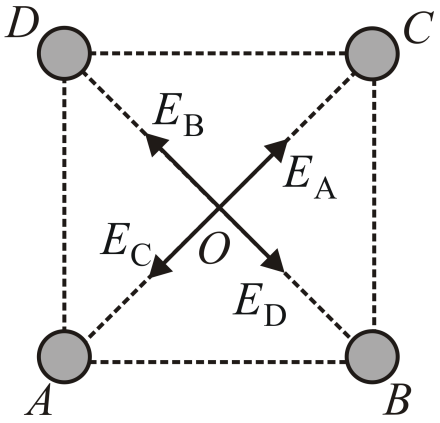
In the case of a closed-loop, the initial and final position of the object is the same. The net displacement of the object is zero. Thus, the net work done by a conservative force in a closed loop is zero.

In a constant electrostatic field, the magnitude of the electric field $(E)$is constant at every point inside the field. When a charged particle is moving from one point to another in the presence of an electrostatic field, the

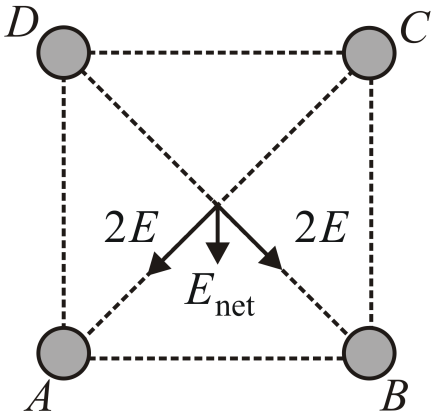
force (F) acting on the particle will be constant since $F = qE$. Thus, this is a conservative force, and work done will be independent of the path connecting P to Q.

Thus, both statements are true, but statement 2 is not the correct explanation of statement 1.

Q9



Let the electric fields generated at the center of the square due to the charge present at the vertices of the square be $\overrightarrow{E_A}$, $\overrightarrow{E_B}$, $\overrightarrow{E_C}$ and $\overrightarrow{E_D}$. Since the magnitude of the electric field is proportional to the magnitude of the charge producing it, let $\left|\overrightarrow{E_A}\right| = E$, $\left|\overrightarrow{E_B}\right| = 2E$, $\left|\overrightarrow{E_C}\right| = 3E$ and $\left|\overrightarrow{E_D}\right| = 4E$.



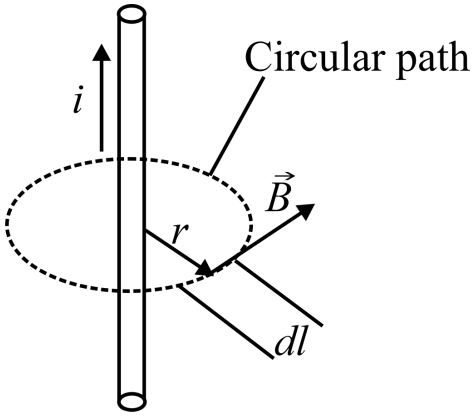
We have,

$\left|\overrightarrow{E_c} + \overrightarrow{E_A}\right| = 2E$ in the direction CA.

$\left|\overrightarrow{E_D}+\overrightarrow{E_B}\right|=2E$ in the direction DB.

These two vectors of equal magnitude at an angle of 90° add up to give the net electric field at the center due to all four charges. The resultant electric field is clearly in the downward direction, or in the direction along CB.

Q10



Consider a long current-carrying wire i, and B is magnetic field at any distance r from the wire.

From Ampere's circuital law,

$\oint \overrightarrow{B}.d\overrightarrow{l}=\mu_0 i$

$\Rightarrow B(2\pi r)=\mu_0 i$

$\Rightarrow B=\frac{\mu_0 i}{2\pi r}$

$\Rightarrow B\propto \frac{1}{r}$.

So, the graph between magnetic field B and distance r will be a rectangular hyperbola.

Q11

Given that the time delay between generation of a wave and the reception of its echo $(T)=77.0$ s.

The generated wave moves through water and strikes the objects under water. After striking with objects, these waves are reflected back.

$\therefore$ Time delay between generation of a wave and the reception of its echo $=$ $2x$ time taken by the wave to reach the enemy submarine

$\therefore$ Time taken by the wave to reach the enemy submarine,

$t = \frac{T}{2} = \frac{77.0}{2}$ s

$= 38.5$ s

Speed of sound in water $(v) = 1450$ m s^{-1}

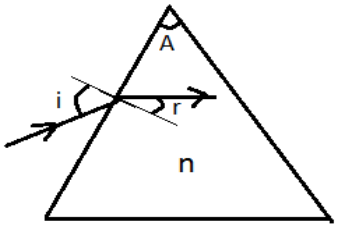
$\therefore$ Distance of enemy submarine $(D) = v \times t$

$= 1450 \times 38.5$

$= 55825$ m

$= 55.825$ km

Q12



at angle of minimum deviation

$\angle i = \angle e \; and \; \angle r = \frac{A}{2}$

$\therefore \partial = i + e - A$

$at \;\; minimum \; deviation$

$\partial_{min} = 2i - A$

$i = \frac{\partial_{min} + A}{2}$

by snell`s law

$n = \frac{\sin\ i}{\sin\ r} \qquad given\ that\ n = cot\left(\frac{A}{2}\right)$

$\Rightarrow cot\left(\frac{A}{2}\right) = \frac{\sin\ \left(\frac{\partial_m + A}{2}\right)}{\sin\frac{A}{2}}$

$\Rightarrow \frac{\cos\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\ \left(\frac{\partial_m + A}{2}\right)}{\sin\frac{A}{2}}$

$\Rightarrow \cos\left(\frac{A}{2}\right) = \sin\ \left(\frac{\partial_m + A}{2}\right)$

$or\ \sin\left(90^\circ - \frac{A}{2}\right) = \sin\left(\frac{A+\delta m}{2}\right)$

$or\ \ 90^\circ - \frac{A}{2} = \ \left(\frac{A+\delta m}{2}\right)$

$or\ \ 180^\circ - A = A + \ \delta m$

$\delta m = 180^\circ - 2A = \pi - 2A$

Q13

The separation between the objective and the eye piece = Length of the telescope, tube $L = f_0 + f_e$

Here, $f_0 = 144\ \text{cm} = 1.44\ \text{m},\ \ f_e = 6.0\text{cm} = 0.06\ \text{m}$

$\therefore\ L = 1.44 + 0.06 = 1.5\ \text{m}$

Q14

The de-Broglie wavelength λ associated with same potential V is

$\lambda = \frac{h}{\sqrt{2meV}}$

$\therefore \lambda\ \alpha \frac{1}{\sqrt{m}}$ [For same potential]

Since alpha particle is $_2^4\text{He}$ and proton is $_1^1\text{H}$

$\therefore$ mass of alpha particle = 4 times mass of proton

$\therefore \lambda_{\text{proton}} \alpha \frac{1}{\sqrt{m}}$ and $\lambda_{\text{alpha}} \alpha \frac{1}{\sqrt{4m}}$

$\therefore \frac{\lambda_{\text{proton}}}{\lambda_{\text{alpha}}} = \frac{1}{\sqrt{m}} \times \frac{\sqrt{4m}}{1} = \frac{2}{1}$

Therefore $\lambda_{\text{proton}} : \lambda_{\text{alpha}} = 2 : 1$

Q15

Kinetic energy of a thermal neutron can be neglected; even for a temperature of $10^6 K$, the thermal energy is only 130 eV. The Q value is given by the equation

$$Q = \left\{M\left(^{235}U\right) + m(n) - \left[M\left(^{99}Zr\right) + M\left(^{134}Te\right) + 3m(n)\right]\right\}c^2$$

$$= [235.0439 - 98.9165 - 133.9115 - 2(1.0087)]u$$

$$= 0.1985 \times 931.5 = 185\ MeV\ .$$

Q16

The input resistance r_{i} of a transistor is in the range of $1 - 10\ \text{k}\Omega$.

The output resistance r_0 of a transistor is in the order of $100\ \text{k}\Omega$.

So, the ratio of output resistance and input resistance,

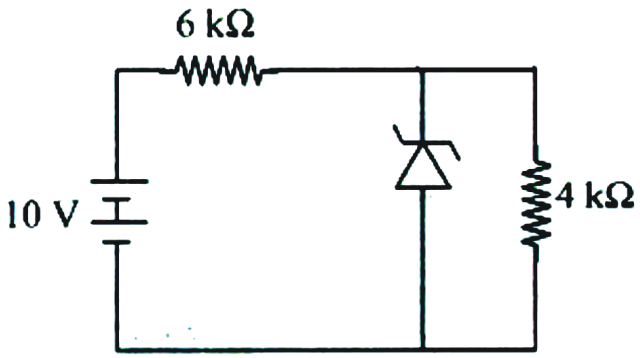
$$R = \frac{r_0}{r_i} = \frac{100\ \text{k}\Omega}{(1-10)\ \text{k}\Omega}$$

$$\Rightarrow\ R \sim 10^2 - 10^3$$

Q17

When $p - n$ junction diode is forward biased, both the depletion region and barrier height are reduced.

Q18



Given, break down voltage is 6V

Then, potential difference across load $4\text{k}\Omega$ will be 6V.

Then potential difference across resistance $6\text{k}\Omega = 10 - 6 = 4\text{V}$

So, current in given resistance is given by Ohm's Law

From $I = \frac{V}{R}$

$= \frac{4}{6000} = \frac{2}{3}\text{mA}$

Q19

The modulated signal (the message is superimposed with high frequency carrier wave) or sender signal is demodulated or reconstruction by a receiver in which a high-frequency carrier wave is knocked out from our information. The reconstruction is always required to get our original message or data.

Q20

Given, wavelength $\lambda_1 = 1\text{ m}$ and $\lambda_2 = 1.01\text{ m}$ and beats per second is $\Delta f = \frac{10}{3}$

Beats is the difference of frequency, then

$\Delta f = f_1 - f_2 \ \ldots(1)$,

where frequency of a wave is $f = \frac{v}{\lambda}$, here v is speed of sound wave.

From equation (1),

$\Delta f = \frac{v}{\lambda_1} - \frac{v}{\lambda_2}$

$\therefore \Delta f = v\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)$

Substituting the values in above equation,

$\frac{10}{3} = v\left(\frac{1}{1} - \frac{1}{1.01}\right)$
$\frac{10}{3} = v\left(\frac{1.01-1}{1\times 1.01}\right)$
$\frac{10}{3} = v\left(\frac{0.01}{1.01}\right)$
$v = 336.66\ \text{ms}^{-1} \simeq 336.67\ \text{ms}^{-1}$

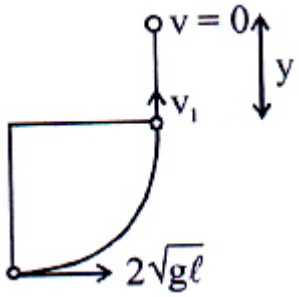
Q21

By energy conservation
$\frac{1}{2}m\left(2\sqrt{gl}\right)^2 = mgh$

$h = 2l$

$y = l$

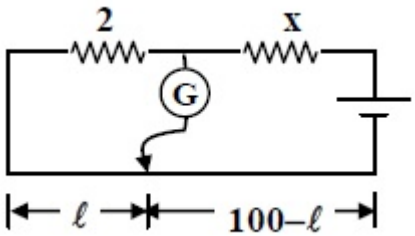
$\text{distance} = \frac{\pi\ell}{2} + \ell = \frac{14}{9}\left[\frac{11}{7} + 1\right] = 4\text{m}$



Q22

Unknown resistor $R > 2\Omega$

$\therefore$ its balencing length is 100 - $l > l$



It form Wheatstone network, then balencing condition for Wheatstone bridge is

Applying $\frac{P}{Q} = \frac{R}{S}$

We have $\frac{2}{X} = \frac{l}{100-l}$ ..... (i)

On intercharging balance point must shift to right as R>2Ω

$\frac{X}{2} = \frac{l+20}{100-l-20}$ ..... (ii)

From (i) and (ii)

$l^2 + 20l = (100 - l)(80 - l)$

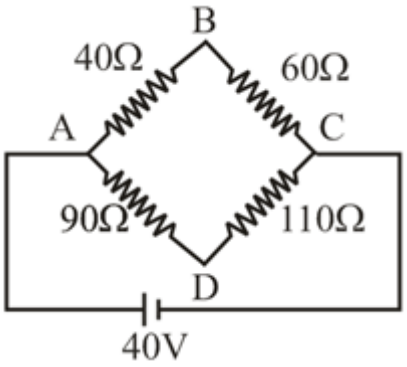
$200l = 8000$

$l = 40\ cm;\ R = 2\Omega$

$For\ 60\ cm,\ resis\tan ce\ will\ be\ 3\Omega$

$\therefore\ \mathrm{X} = 3\ \Omega$

Q23



From KVL

$\mathrm{V_B} - 60\left(\frac{40}{100}\right) + 110\left(\frac{40}{200}\right) = \mathrm{V_D}$

$\mathrm{V_B} - \mathrm{V_D} = 2$

Q24

$R = 50\ \Omega$

$X_L = \omega L = 50\ \Omega$

$X_L = \frac{1}{\omega C} = 100\ \Omega$

$\Rightarrow Z = \sqrt{R^2 + (X_L - X_C)^2}$

$Z = 50\sqrt{2}\,\Omega$

$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{200/\sqrt{2}}{50\sqrt{2}} = 2\ \mathrm{A}$

Q25

$I\omega_0 = 2I\omega$ ($M.I.$ of one disc)

$\therefore K = \frac{1}{2}2I\omega^2$

$= \frac{1}{2}\left(\frac{1}{2}I\omega_0^2\right)$

The ratio of kinetic energy before and after collision is 2

Q26

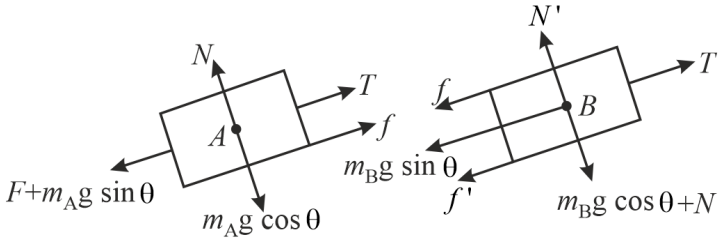
From momentum conservation, Velocity of B just after impact $= \frac{v}{2}$

Angular velocity, $\omega = \frac{v}{2l}$

So, $n = 2$

Q27

FBD of Block A and B.



By Newton's II law of motion,

For Block A,

$m_A g \sin\theta + F - T - f = m_A a \ \ldots (\mathrm{i})$

Kinetic frictional force on A, $\quad f = \mu_k m_A g \cos\theta \ \ldots (\mathrm{ii})$

For Block B,

$T - m_B g \sin\theta - f - f' = m_B a \ \ldots (\mathrm{iii})$

Kinetic frictional force on B, $\quad f' = \mu_k . N' \ldots (\mathrm{iv})$

$= \mu_k (m_A g \cos\theta + m_B g \cos\theta)$

Solving above equations, then

$a_A = 5.2 \ \mathrm{ms}^{-2}$

Q28

For ideal gas :

$\Delta \mathrm{U} = \mathrm{nC_V}[\mathrm{T_2} - \mathrm{T_1}]$

$\Rightarrow \quad 5000 = 4 \times \mathrm{C_V}[500 - 300]$

$\Rightarrow \mathrm{C_V} = \frac{5000}{800} = 6.25 \ \mathrm{J \ mole^{-1} K^{-1}}$

Q29

$\sin\theta = \frac{L}{r} = \frac{1}{\frac{mv}{Bq}}$

$\therefore \quad v \propto \frac{1}{\sin\theta} \; or \; \frac{v}{v_0} = \frac{\sin 60^\circ}{\sin 30^\circ}$

$\therefore \quad v = \sqrt{3}v_0$

Q30

In series LCR, current is maximum at resonance.

$\therefore$ Resonant frequency, $\omega = \frac{1}{\sqrt{LC}}$

$\therefore \; \omega^2 = \frac{1}{LC}$ or, $L = \frac{1}{\omega^2 C}$

Given, $\omega = 1000 \text{ s}^{-1}$ and $C = 10 \; \mu\text{F}$

$\therefore \; L = \frac{1}{1000\times1000\times10\times10^{-6}} = 0.1 \text{ H} = 100 \text{ mH} \Rightarrow n = 5$

Q31

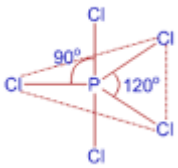
Non – stoichiometric Schottky defect is a type of point defect. This defect is forms when oppositely charged ion leave their lattice site creating vacancies, thus lowers the density of crystal.

Q32

For H-atom E belongs to visible region when $\text{n}_1 = 2 \text{and } \text{n}_2 = 3$. B, C and D lines belongs to Infrared region.

Q33

On orbital overlap between phosphorus and chlorine, five sp^3 d-p sigma covalent bonds are formed. This statement is true, as PCl_5 carries five sigma bonds and all these sigma bonds are used for hybridisation according to the valence shell electron pair repulsion theory. Structure of PCl_5 is shown below:



And we can see here, that axial bonds and equatorial bonds, both are not of the same length.

Q34

$\mathrm{pH}=\mathrm{pk}_{\mathrm{a}}+\log\left[\frac{\text{Salt}}{\text{Acid}}\right]\ (\because [\text{Salt}]=[\text{Anion}])$

$\Rightarrow 6=5+\log\ \frac{\text{Salt}}{\text{Acid}}$

$\Rightarrow 1=\log\ \frac{\text{Salt}}{\text{Acid}}$

$\Rightarrow \log 10=\log\ \frac{\text{Salt}}{\text{Acid}}$

$\frac{\text{Salt}}{\text{acid}}=\frac{10}{1}$

Q35

In manufacture of $\mathrm{H_2\,SO_4}$ (contact process), $\mathrm{V_2O_5}$ is used as a catalyst. Ni catalysts enables the hydrogenation of fats. $\mathrm{CuCl_2}$ is used as catalyst in Deacon's process. $\mathrm{ZSM-5}$ used as catalyst in cracking of hydrocarbons.

Q36

$\mathrm{He} \Rightarrow$ Inert gas

$\mathrm{Cl} \Rightarrow$ Electron gain enthalpy is highest

$\mathrm{Ca} \Rightarrow$ Most electropositive metal

$\mathrm{Li} \Rightarrow$ Strong reducing agent

He has the electronic configuration of $1s^2$. It is completely filled electronic configuration, due to this, it has the highest ionisation enthalpy.

$\mathrm{Cl} \longrightarrow \mathrm{Cl}^-$ gives the highest $\Delta_{\mathrm{eg}}\mathrm{H}$ as Cl^- after getting one electron, it will achieve the stable electronic configuration $18[\mathrm{Ar}]$. Due to this, large amount of energy is released during this process.

$_{20}\mathrm{Ca} \underset{-2\mathrm{e}^-}{\longrightarrow} \mathrm{Ca}^{2+}$

Alkaline earth metal and present in the fourth period, the outermost electron is far away from the nucleus. As a result, less energy is required to remove the electron and easily donate the electron, hence, shows maximum metallic nature.

E^{o}_{red} of Li is -3.07 V, a strong reducing agent.

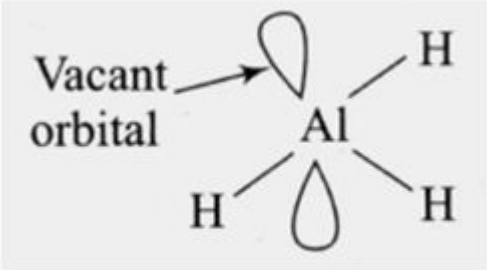
It has very high hydration energy due to smaller in size. The hydration energy is directly proportional to the charge density of the ion.

Q37

On being preferrentially wetted by oil, the ore particles rise to the surface in the form of froth and from there we can separate them.

Q38

AlH_3 is an electron deficient hydride.



It is generally formed by the group 13 element, and have lesser number of electrons than that required for writing its Lewis structure. Being electron deficient, this hydride generally behaves as a Lewis acid, which act as electron acceptor. This is a polynuclear hydride. According to the octet rule, each element tends to completely fill it's outermost shell with $8e^-$ in it. The electronic configuration of aluminium is $2, 8, 3$, and it still needs 5 more electrons to complete its octet. Al has 3 valence electrons to complete its octet, while each hydrogen has one valence electron.

Q39

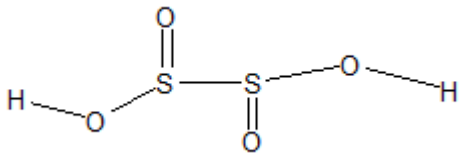
The stability of the oxide of alkali metals depends upon the comparability of size of cation and anion.

Therefore the main oxide of alkali metals formed on excess of air are as follows

Li	Li_2O
Na	Na_2O_2
K	KO_2
Rb	RbO_2
Cs	CsO_2

Q40

Hyposulphurous acid contains the $S-S$ bond linkage in the structure having the chemical formula $H_2S_2O_4$.



Q41

$\left[Fe(H_2O)_6\right]^{3+}$

$Fe^{+2} = 3d^5\left(t_{2g}^{1,1,1}e_g^{1,1}\right)$

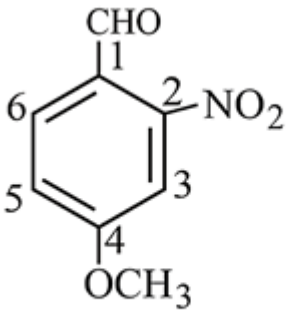
So C.F.S.E is $=[-0.4\times 3+0.6\times 2]\Delta_0 = 0$

Q42

In Calcination and roasting CO_2 and SO_2 are released which are responsible for Global warming and acid rain.

Q43

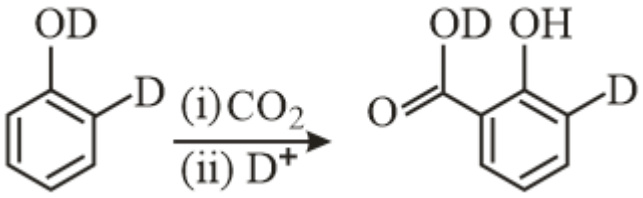
The -CHO functional group is the highest priority functional group. The carbons of the benzene ring are numbered accordingly.



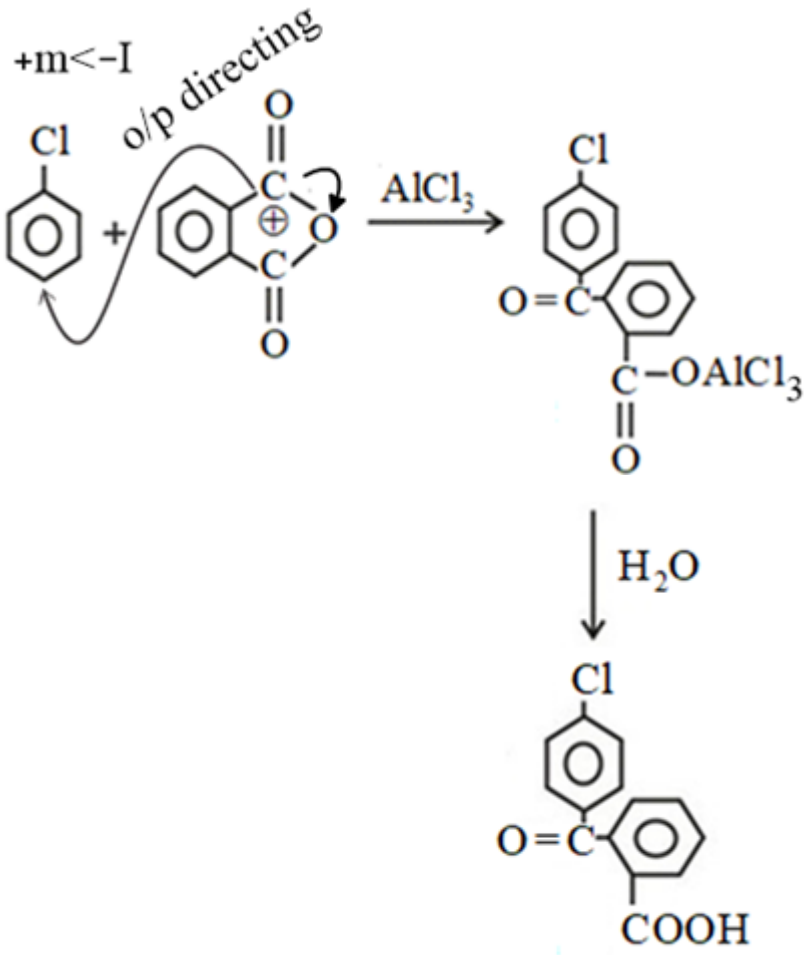
4-methoxy-2-nitrobenzaldehyde

Q44

In the first step of the reaction, NaOH is given which is a strong base, abstracts the hydrogen and phenoxide ion is formed, which is more reactive than phenol. It further undergoes electrophilic substitution reaction with CO_2 to form salicylic acid. But further reaction with D^+ abstracts the hydrogen of acid to form deuterated acid. It follows the mechanism of Kolbe's reaction.

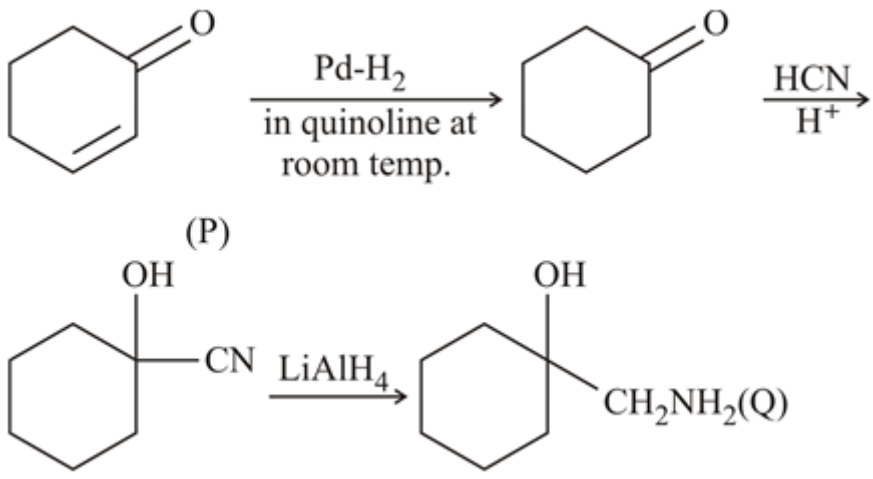


Q45

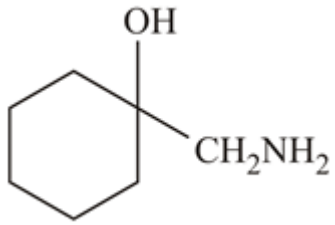


This reaction is Friedel-craft-acylation, $-Cl$ group is an ortho and para directing.

Q46



$\therefore$ P is $Pd-H_2$, in quinoline at room temp.,



Q is

Q47

Explanation :- aniline is more basic than acetamide because in acetamide, lone pair of nitrogen is delocalised to more electronegative element oxygen.

In Aniline lone pair of nitrogen delocalised over benzene ring.

Q48

High density polythene: It is formed when addition polymerisation of ethene takes place in a hydrocarbon solvent in the presence of a catalyst such as triethylaluminium and titanium tetrachloride (Ziegler-Natta catalyst) at a temperature of 333 K to 343 K and under a pressure of 6-7 atmospheres. High density polythene (HDP) thus produced, consists of linear molecules and has a high density due to close packing. It is also chemically inert and more tougher and harder. It is used for manufacturing buckets, dustbins, bottles, pipes, etc.

Q49

In double stranded DNA, the molar amount of adenine equals that of thymine, and the molar amount of guanine equals that of cytosine. Therefore, A + G = T + C, but A + T does not equal G + C.

Q50

Biodegradable detergents are defined as the type of detergent that has a straight hydrocarbon chain. These detergents are known as biodegradable as they are destroyed by the bacteria.

Branched alkyl groups are not dissociated easily and hence, such compounds are non-biodegradable.

Q51

$\frac{R_{H_2}}{R_{H.C}} = \sqrt{\frac{M_{H.C}}{2}} = 3\sqrt{3}$

$M_{H.C} = 54\ g/mol$

$M_{H.C} = 12n + (2n - 2) = 54$

$n = 4$

Q52

$PdV + VdP = nRdT$ ...(i)

along $AB \rightarrow PT = constant$

$PdT + TdP = 0$

$dP = -\frac{P}{T}dT$

Substitute in (i)

$PdV + V\left(-\frac{P}{T}dT\right) = nRdT$

$PdV = nRdT + \frac{PV}{T}dT = 2nRdT$

$W_{AB} = -\int_{V_1}^{V_2} PdV = -R\int_{2T_1}^{T_1} 2ndT = -2nR[T_1 - 2T_1]$

$= -2 \times 2 \times R[300 - 600]$

$= 1200\ R = (150\ R)x$

$\therefore\ x = 8$

Q53

$A + B \rightleftharpoons 2C$		
1	1	1
-x	-x	2x
1-x	1-x	1 + 2x

$$\mathrm{K}=\frac{[\mathrm{C}]_{\mathrm{eq}}^{2}}{[\mathrm{A}]_{\mathrm{eq}}[\mathrm{B}]_{\mathrm{eq}}}=\frac{(1+2\mathrm{x})^{2}}{(1-\mathrm{x})(1-\mathrm{x})}$$

$$100=\left(\frac{1+2\mathrm{x}}{1-\mathrm{x}}\right)^{2}$$

$$\left(\frac{1+2\mathrm{x}}{1-\mathrm{x}}\right)=10$$

$$\mathrm{x}=\frac{3}{4}$$

$$[\mathrm{C}]\mathrm{e}_{\mathrm{q.}}=1+2\mathrm{x}$$

$$=1+2\left(\frac{3}{4}\right)$$

$$=2.5\ \mathrm{M}$$

$$=25\times10^{-1}\ \mathrm{M}$$

Q54

$$\kappa=\frac{1}{R}\cdot G^{*}$$

For same conductivity cell, G^{*} is constant and hence $\kappa\cdot R.=$ constant.

$$\therefore 0.14\times4.19=\kappa\times1.03$$

or, $=\frac{0.14\times4.19}{1.03}$

$$=0.5695\,\mathrm{Sm}^{-1}$$

$$=56.95\times10^{-2}\,\mathrm{Sm}^{-1}\approx57\times10^{-2}\,\mathrm{Sm}^{-1}$$

Q55

From question

$\frac{2k_1}{2k_1+3k_2}=0.5$ and $k_1+k_2=\frac{0.693}{1.386\times10^{2}}=5\times10^{-3}h^{-1}$

Solving $\frac{k_1}{k_2}=\frac{2}{3}$ and $k_1=2\times10^{-3}h^{-1}$ and $k_2 = 3 \times 10^{-3} h^{-1}$

$t_{1/2(A\rightarrow B)}=\frac{0.693}{k_2}=\frac{0.693}{3\times10^{-3}} = 231 h$

Q56

$8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^{\ominus}$

Q57

Complexes (i), (iii),(iv) and (v) are optically inactive due to the presence of plane of symmetry.

Q58

Considering 100 g of solid, mass of anhydrous salt $= 56.25\ g$ and mass of water $= 100 - 56.25$

$43.75\ g$

Moles of H_2O in 288 g of solid $= \frac{43.75}{100} \times 288$

$= 126\ g$

$\therefore$ Moles of $H_2O = \frac{126}{18} = 7\ mol$

$\therefore n = 7$

Q59

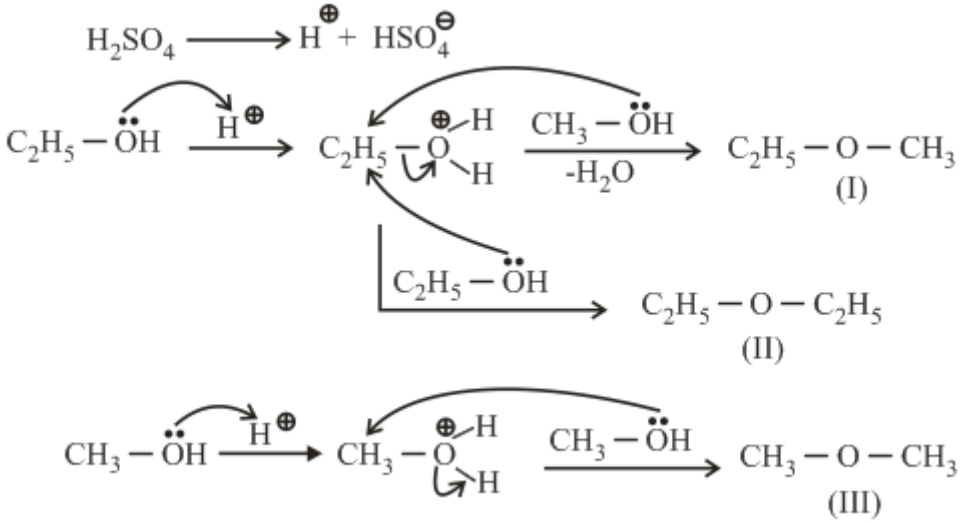
When alcohol is treated with concentrated H_2SO_4, ethers are formed by dehydration of alcohols.

Same alcohol form symmetric ether.

Different alcohol form unsymmetric ether.

The reaction follows SN^2 mechanism.

Mechanism:



Hence, three different ethers are formed.

Q60

$n_{eq}\, Fe^{2+} = n_{eq}\, Cr_2\, O_7^{2-}$

or $\left(\frac{15 \times M_{Fe^{2+}}}{1000}\right) \times 1 = \left(\frac{20 \times 0.03}{1000}\right) \times 6$

$\therefore M_{Fe^{2+}} = 0.24M = 24 \times 10^{-2}M$

Q61

Since, $\alpha^2 - 2\alpha + 3 = 0$

$\Rightarrow \alpha^3 - 3\alpha^2 + 5\alpha - 2 = \alpha(\alpha^2 - 2\alpha + 3) - (\alpha^2 - 2\alpha + 3) + 1 = 1$

And $\beta^2 - 2\beta + 3 = 0$

$\Rightarrow \beta^3 - \beta^2 + \beta + 5 = \beta(\beta^2 - 2\beta + 3) + (\beta^2 - 2\beta + 3) + 2 = 2$

So, the required equation is

$x^2 - (2 + 1)x + 2.1 = 0$

$\Rightarrow x^2 - 3x + 2 = 0$

Sum of roots $= 3$

Q62

Here $\frac{z_1+z_2+z_3}{3} = z_0 \Rightarrow z_3 = 4 + 3z_0$

Therefore, centroid coincides with the circumcentre

$\Rightarrow$ Triangle is equilateral $\Rightarrow |z_1 - z_2| = 4$

Clearly, z_3 either lie in the second or third quadrant

So the centre z_0 also lies in the second or third quadrant.

$\therefore z_0$ can be $= -2 + \frac{2}{\sqrt{3}}\mathrm{i}, -2 - \frac{2}{\sqrt{3}}\mathrm{i}$

$\Rightarrow \arg(z_0) = \frac{5\pi}{6}, \frac{7\pi}{6}$

Q63

$\because P_n = \left[\frac{3i+j}{4^{2n}}\right] = \frac{1}{4^{2n}}[3i + j]$

$4^n P_n = \frac{1}{4^n}[3i + j]$

$T_r\left(4P_1 + 4^2P_2 \ldots 4^nP_n\right) = T_r(4P_1) + T_r\left(4^2P_2\right) \ldots . + T_r(4^nP_n)$

$= \frac{24}{4} + \frac{24}{4^2} \ldots \ldots \ldots \frac{24}{4^n}$

$\lim\limits_{n\to\infty}\left(T_r\left(4P_1 + 4^2P_2 \ldots . + 4^nP_n\right)\right)$

$\lim\limits_{n\to\infty}\left(\frac{24}{4} + \frac{24}{4^2} \ldots \ldots . \frac{24}{4^n}\right)$

$= \frac{6}{1-\frac{1}{4}} = 8$

Q64

$T_k = \frac{5+(k-1)4}{(3+(k-1)4)^2(7+(k-1)4)^2}$

$= \frac{4k+1}{(4k-1)^2(4k+3)^2}$

$= \frac{1}{8}\left\{\frac{1}{(4k-1)^2} - \frac{1}{(4k+3)^2}\right\}$

$S_n = T_1 + T_2 + T_3 + \ldots . + T_n$

$= \frac{1}{8}\left\{\frac{1}{3^2} - \frac{1}{7^2} + \frac{1}{7^2} - \frac{1}{11^2} + \frac{1}{11^2} - \frac{1}{15^2} + \ldots . + \frac{1}{(4n-1)^2} - \frac{1}{(4n+3)^2}\right\}$

$$= \frac{1}{8}\left\{\frac{1}{3^2} - \frac{1}{(4n+3)^2}\right\}$$

$$S_\infty = \lim_{n\to\infty} S_n = \frac{1}{8}\left\{\frac{1}{9} - 0\right\} = \frac{1}{72}$$

Q65

$$\lim_{x\to 0} \frac{x^{6000} - (\sin x)^{6000}}{x^2 \frac{(\sin x)^{6000}}{x^{6000}} . x^{6000}} \qquad \left(\frac{0}{0}\right)$$

$$\lim_{x\to 0} \frac{x^{6000} - (\sin x)^{6000}}{x^{6002}}$$

$$= -\lim_{x\to 0} \frac{(\sin x)^{6000} - x^{6000}}{x^{6002}}$$

$$= -\lim_{x\to 0} \frac{\left(\frac{\sin x}{x}\right)^{6000} - 1}{x^2}$$

$$= -\lim_{x\to 0} \frac{6000\left(\frac{\sin x}{x}\right)^{5999}\left(\frac{x\cos x - \sin x}{x^2}\right)}{2x}$$ (By L-Hospital Rule)

$$= -\frac{6000}{2}\lim_{x\to 0} \frac{x\cos x - \sin x}{x^3}$$

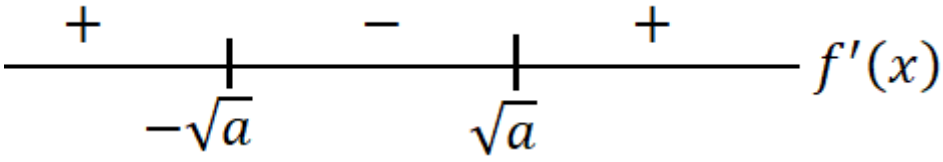
$$= -3000 \lim_{x\to 0} \frac{\cos x - x\ \sin x - \cos x}{3x^2}$$

$$= 1000$$

Q66

$$f'(x) = 3x^2 - 3a$$

$$= 3\left(x - \sqrt{a}\right)\left(x + \sqrt{a}\right)$$



$\therefore f(x)$ has a local minimum at $x = \sqrt{a}$

$\Rightarrow \sqrt{a} \geq 4$

$\Rightarrow a \geq 16$

$\therefore$ 'a' can be 16,17,18

Q67

$$\sin x + \cos x = \sqrt{2}\cos\left(x - \frac{\pi}{4}\right) \in [1, \sqrt{2}] \forall x \in [0, 1]$$

Hence, as $x \in [0, 1]$

$$10x^{\sqrt{2}} \leq f(x) \leq 10x$$

With equality holding if and only if $x = 0$ or 1

As equality holds only for finitely many points, the inequalities become on strict integrating on all sides.

Hence, $4 < 10(\sqrt{2} - 1) < \int_0^1 f(x)dx < 5 \Rightarrow \left[\int_0^1 f(x)dx\right] = 4$

Q68

$f'(x) = \tan^2 x + K$ where $K = \int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} f(x)dx$

$$f(x) = \tan x - x + Kx + C$$
$$f\left(\frac{\pi}{4}\right) = 1 - \frac{\pi}{4} + \frac{K\pi}{4} + C = \frac{-\pi}{4}$$
$$C + 1 = \frac{-K\pi}{4}$$
$$f(x) = \tan x - x + Kx - \frac{K\pi}{4} - 1$$

Now,

$K = \int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} (\underbrace{\tan x - x + Kx}_{\text{odd function}} - \frac{K\pi}{4} - 1)dx = \frac{-\pi}{2} - \frac{K\pi^2}{8}$

Hence, $K = \frac{-4\pi}{8+\pi^2}$

$$\therefore \frac{8 + \pi^2}{\pi} \cdot \int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} f(x)dx = -4 \equiv m \Rightarrow m^2 = 16$$

Q69

Given equation is $x\cos ydy+\sin ydx=xdx$

Or $d(x\sin y)=xdx$

Integrating, we get $\int d(x\sin y)=\int xdx$

$\Rightarrow x\sin y=\frac{x^2}{2}+C$

Since, it passes through $(0,0)$

$C=0$

$\Rightarrow x\sin y=\frac{x^2}{2}$

Q70

Parameter corresponding to P is $t_1=3$

Hence, the parameter corresponding to Q is $t_2=-t_1-\frac{2}{t_1}=-3-\frac{2}{3}=-\frac{11}{3}$

Let, $S(t_3)$ be the point where another tangent from R touches the parabola

Now, if the tangent at $Q(t_2)$ and $S(t_3)$ meet on the directrix at R then $t_2t_3=-1$

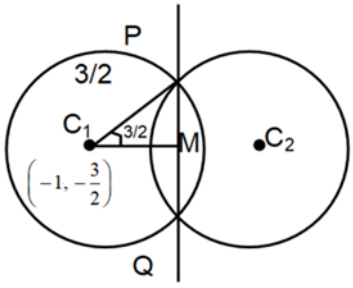
$\Rightarrow t_3=\frac{3}{11}$

So the equation of the tangent at S is $\frac{3}{11}y=x+\frac{9}{121}$

Hence, the slope of the tangent at S is $\frac{11}{3}$

Q71

Equation of the common chord is $2x+1=0$



$C_1M=\left|\frac{-2+1}{2}\right|=\frac{1}{2}$

$PM = \sqrt{\frac{9}{4} - \frac{1}{4}} = \sqrt{2}$

Length of the common chord $= 2\sqrt{2}$

Hence, the perimeter of $\Delta C_1PQ = \frac{3}{2} + \frac{3}{2} + 2\sqrt{2}$

$= 3 + 2\sqrt{2}$ units

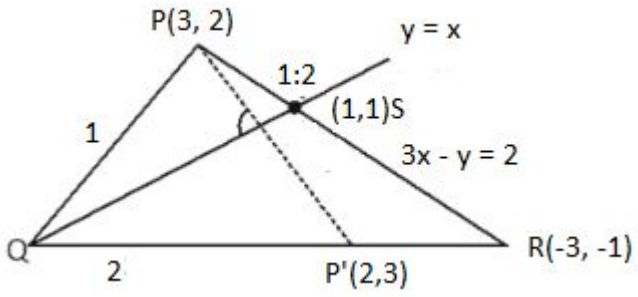
Q72

Plane P_1 is parallel to line AB

$\Rightarrow$ Shortest distance $=$ Perpendicular distance of any point on line to plane $P_1 = 0$

$= \left|\frac{0+9+4+1}{\sqrt{9+1}}\right| = \frac{14}{\sqrt{10}} = 7\sqrt{\frac{2}{5}}$ units

Q73



Point of intersection of $y = x$ & $3x - y = 2$ is $S(1,1)$

Now, $PS : SR = PR : RQ = 1 : 2$

Hence, using section formula $R = (-3, -1)$

Let, reflection of P about the line $y = x$ is $P^{'}$

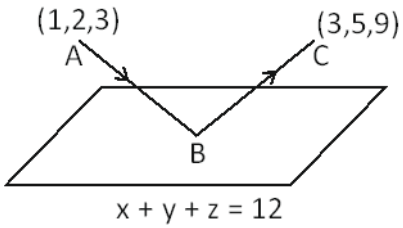
$\Rightarrow P^{'} = (2,3)$ which lies on the side QR

$\Rightarrow$ equation of QR is $4x - 5y + 7 = 0$

Point of intersection of QR & $y = x$ is

$(7,7) = Q$

Q74



A plane which contains the incident ray and the reflected ray must contain the points A, B, C and should be perpendicular to the given plane.

So, equation of the plane is $\begin{vmatrix} x-1 & y-2 & z-3 \\ 1 & 1 & 1 \\ 2 & 3 & 6 \end{vmatrix} = 0$

$\Rightarrow (x-1)(3)-(y-2)(4)+(z-3)(3-2)=0$

$\Rightarrow 3x-3-4y+8+z-3=0$

$\Rightarrow 3x-4y+z+2=0$

Hence, the distance of the plane from $(0,0,0)$ is $\left|\frac{0-0+0+2}{\sqrt{9+16+1}}\right| = \frac{2}{\sqrt{26}}$ units

Q75

Given, $\vec{r} \times \vec{a} = \vec{b}$

Taking right cross product with $\vec{c}$, we get,

$\left(\vec{r} \times \vec{a}\right) \times \vec{c} = \vec{b} \times \vec{c} \Rightarrow \left(\vec{r} \cdot \vec{c}\right)\vec{a} - \left(\vec{a} \cdot \vec{c}\right)\vec{r} = \vec{b} \times \vec{c}$

$\Rightarrow 3\vec{a} - \vec{r} = \vec{b} \times \vec{c} \Rightarrow \vec{r} = 3\left(2\hat{i}+3\hat{j}+4\hat{k}\right) - \left(\hat{i}+2\hat{j}+3\hat{k}\right)$

$\Rightarrow \vec{r} = 5\hat{i}+7\hat{j}+9\hat{k} \Rightarrow \left|\vec{r}\right| = \sqrt{155}$

Q76

For A,

$|x-3| = 1 \Rightarrow x = 2,4$ or

$x^2 - 5x + 6 = 0 \Rightarrow x = 2,3$ but $x \neq 3$

$\therefore A = \{2,4\}$

For B,

$B = \{4, 5, 6\}$

$n(A \times B) = 6$

$\therefore$ number of subsets $= 2^6$

Q77

$\bar{x}_{\text{old}} = 10 = \frac{\Sigma x_i}{10} \Rightarrow \Sigma x_{i\text{old}} = 100$

$\Sigma x_{i\text{new}} = 100 - 5 + 15 = 110$

$\bar{x}_{\text{new}} = \frac{110}{10} = 11$

$\text{Var}_{\text{old}} = 5 = \frac{\Sigma x_{i\text{old}}^2}{10} - (\bar{x}_{\text{old}})^2$

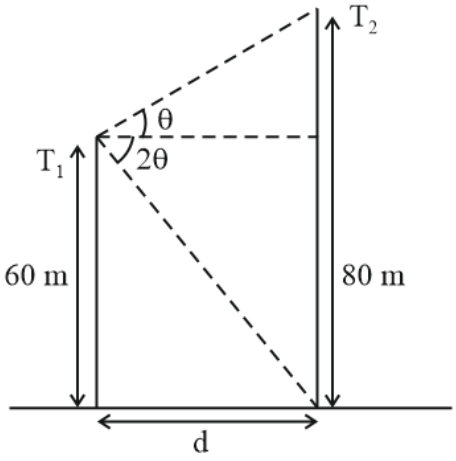
$\Sigma x_{i\text{old}}^2 = 1050$

$\Sigma x_{i\,\text{new}}^2 = 1050 - 25 + 225$

$= 1250$

$\text{Var}_{\text{new}} = \frac{1250}{10} - (11)^2 = 125 - 121 = 4$

Q78



Let, the width of the road the road is d.

If the angle of the elevation is θ, then

$\tan\theta = \frac{20}{d}$, here 20 is the difference between the heights of T_1 and T_2.

Given that, the angle of depression is twice of the angle of elevation.

$\tan 2\theta = \frac{60}{d}$

We know, $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

$\therefore \frac{60}{d} = \frac{40/d}{1-(400/d^2)}$

$\Rightarrow \frac{400}{d^2} = \frac{1}{3}$

$d = 20\sqrt{3}$

Q79

$(p \wedge q \wedge r)$ is subset of r, so $(p \wedge q \wedge r)$ is a subset of $(\sim p \vee \sim q \vee r)$. Hence, $(p \wedge q \wedge r) \Rightarrow (\sim p \vee \sim q \vee r)$ is a tautology.

Similarly, $(p \wedge q \wedge r) \Rightarrow ((\sim p \wedge \sim q) \vee r)$ is a tautology because $(p \wedge q \wedge r)$ is a subset of $((\sim p \wedge \sim q) \vee r)$

Similarly, $(p \wedge \sim q \wedge r)$ is a subset of $(\sim p \vee q \vee r)$. Hence, $(p \wedge \sim q \wedge r) \Rightarrow (\sim p \vee q \vee r)$ is a tautology

$(p \wedge q \wedge \sim r) \Rightarrow r$ is false if p is true, q is true and r is false. Hence, $(p \wedge q \wedge \sim r) \Rightarrow r$ is not a tautology

Q80

$(1+x)^n = \sum_{r=0}^{n} C_r x^r$

$\Rightarrow x(1+x)^n = \sum_{r=0}^{n} C_r x^{r+1}$

Differentiating w.r.t. x we get

$xn(1+x)^{n-1} + (1+x)^n = \sum_{r=0}^{n}(r+1)C_r x^r$

Again multiplying both sides by x

$(1+x)^{n-1}(nx+1+x)x = \sum_{r=0}^{n}(r+1)C_r x^{r+1}$

Again differentiating w.r.t. x we get,

$\frac{d}{dx}\left((1+x)^{n-1}\left(nx^2+x^2+x\right)\right) = \sum_{r=0}^{n}\left((r+1)^2\right)C_r x^r$

$\sum_{r=0}^{n}\left((r+1)^2\right)C_r x^r = (1+x)^{n-1}(2nx+2x+1) + \left(nx^2+x^2+x\right)(n-1)(1+x)^{n-2}$

Putting $x = 1$ on both sides, we get,

$\sum_{r=0}^{n}\left((r+1)^2\right)C_r = 2^{n-1}(2n+3) + (n+2)\cdot(n-1)2^{n-2} = 2^{n-2}\left(n^2+5n+4\right)$

Now, $f(x) = x^2+5x+4 = (x+1)(x+4) \Rightarrow \alpha = -1,\ \beta = -4$

Hence, $\alpha^2 + \beta^2 = 17$

Q81

Let, $\vec{\alpha}=\hat{i}-\hat{j}+2\hat{k},\ \vec{\beta}=2\hat{i}-\hat{j}+\hat{k},\ \vec{\gamma}=\hat{i}-2\hat{j}+\hat{k}$

$\vec{a}=\frac{\vec{\gamma}\times\left(\vec{\alpha}\times\vec{\beta}\right)}{\left|\vec{\gamma}\times\left(\vec{\alpha}\times\vec{\beta}\right)\right|}$

$\vec{\gamma}\times\left(\vec{\alpha}\times\vec{\beta}\right)=\left(\vec{\gamma}\cdot\vec{\beta}\right)\vec{\alpha}-\left(\vec{\gamma}\cdot\vec{\alpha}\right)\vec{\beta}$

$=5\vec{\alpha}-5\vec{\beta}$

$=5\left(-\hat{i}+\hat{k}\right)$

$\vec{a}=\frac{-\hat{i}+\hat{k}}{\sqrt{2}}$

Projection of $\vec{a}$ on $\vec{\alpha}$ is $\vec{a}\cdot\widehat{\alpha}=\frac{\vec{a}\cdot\vec{\alpha}}{\left|\vec{\alpha}\right|}=\frac{-1+2}{\sqrt{2}\sqrt{6}}=\frac{1}{\sqrt{12}}$

Q82

$\frac{dy}{dx}=(1+x)+\left(\frac{y-3}{1+x}\right)$

$\frac{dy}{dx}-\frac{1}{(1+x)}y=(1+x)-\frac{3}{(1+x)}$

I.F$=e^{-\int\frac{1}{(1+x)}dx}=\frac{1}{(1+x)}$

$\therefore\frac{d}{dx}\left(\frac{y}{1+x}\right)=x+3(1+x)^{-1}+c$

$\Rightarrow\frac{y}{1+x}=x+3(1+x)^{-1}+c$

$\Rightarrow y=(1+x)\left[x+\frac{3}{(1+x)}+c\right]$, $\therefore$ at $x=2,\ y=0$

$\Rightarrow 0=3(2+1+c)\Rightarrow c=-3$

At $x=3,\ y=3$

Q83

$2\left(\frac{3x}{2}\right)+3\left(\frac{4y}{3}\right)=5$

$\frac{3x}{2}+\frac{3x}{2}+\frac{4y}{3}+\frac{4y}{3}+\frac{4y}{3}=5$

Using $A.M\geq G.M$ for $\frac{3x}{2},\frac{3x}{2},\frac{4y}{3},\frac{4y}{3},\frac{4y}{3}$

$\frac{\frac{3x}{2}+\frac{3x}{2}+\frac{4y}{3}+\frac{4y}{3}+\frac{4y}{3}}{5} \geq \left(\left(\frac{3x}{2}\right)^2\left(\frac{4y}{3}\right)^3\right)^{\frac{1}{5}}$

$\frac{3x+4y}{5} \geq \left(\frac{3^2 4^3}{2^2 3^3}x^2y^3\right)^{\frac{1}{5}}$

$1 \geq \left(\frac{2^4}{3}x^2y^3\right)^{\frac{1}{5}}$

$16x^2y^3 \leq 3$

Maximum value of $16x^2y^3 = 3$

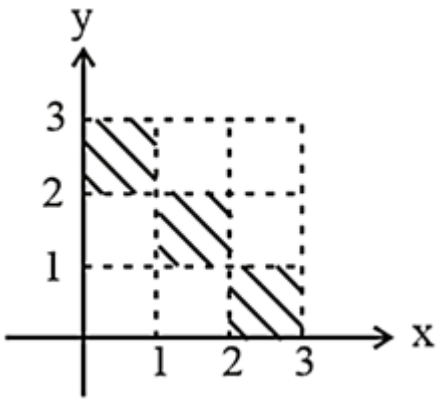
Q84

$\because \sin^{-1}\left(\frac{2\sqrt{15}}{|x|}\right) = \cos^{-1}\sqrt{1-\left(\frac{2\sqrt{15}}{|x|}\right)^2}$ for $0 < \frac{2\sqrt{15}}{|x|} \leq 1 \Rightarrow |x| \geq 2\sqrt{15}$

$\therefore \left(\frac{14}{|x|}\right)^2 = 1-\left(\frac{2\sqrt{15}}{|x|}\right)^2 \Rightarrow |x| = 16 \Rightarrow x = \pm 16$ which satisfies $|x| \geq 2\sqrt{15}$.

Hence, the maximum value of $x = 16$

Q85



$[x]+[y] = 2$ (Let $x, y \geq 0$)

$\Rightarrow [x], [y] = 2,0$ or $1,1$ or $0,2$

If $[x] = 2$ and $[y] = 0$

$\Rightarrow x \in [2,3)$ and $y \in [0,1)$ and so on

Since, the curve is symmetric in the Ist and IIIrd quadrant

Hence, the required area $= 6(1 \times 1) = 6$ sq. units

Q86

$$\begin{aligned} f(x) &= x^4 - 4x^3 - 8x^2 + a \\ f'(x) &= 4\left(x^3 - 3x^2 - 4x\right) \\ &= 4x\left(x^2 - 3x - 4\right) \\ &= 4x(x-4)(x+1) = 0 \text{ at } x = -1, 0, 4 \\ f(-1) &= a - 3 \leq 0, a \leq 3 \\ f(0) &\geq 0 \Rightarrow a \geq 0 \\ a &\in [0, 3] \\ \text{Sum } &= 0 + 1 + 2 + 3 = 6 \end{aligned}$$

Q87

Let, $Z = re^{i\theta}$

$\Rightarrow r^3 e^{i\theta} + 4re^{-i2\theta} = 0$

$\Rightarrow r^2 e^{i5\theta} = -4$

$\Rightarrow r^2 = 4$ and $e^{i5\theta} = -1$

$\Rightarrow r = 2$ and $\theta = \frac{-3\pi}{5}, \frac{-\pi}{5}, \frac{\pi}{5}, \frac{3\pi}{5}, \pi$

$\Rightarrow$ least arg Z is $-\frac{3\pi}{5}$

$\Rightarrow k = 0.6$

Q88

Let, $y = x + h$, where $h \geq 1$ and $z = x + h + k$, where $k \geq 1$.

Then, $x + y + z = 20$

$\Rightarrow x + x + h + x + h + k = 20$

$\Rightarrow 3x + 2h + k = 20.$

When $x = 1 \Rightarrow 2h + k = 17$, then there are 8 ways.

When $x = 2 \Rightarrow 2h + k = 14$, then there are 6 ways.

When $x = 3 \Rightarrow 2h + k = 11$, then there are 5 ways.

When $x = 4 \Rightarrow 2h + k = 8$, then there are 3 ways.

When $x = 5 \Rightarrow 2h + k = 5$, then there are 2 ways.

When $x > 5 \Rightarrow$ not possible.

$\Rightarrow$ Total number of ways $= 24$

Q89

$\ell = \mathrm{n}^2 - \frac{\mathrm{n}(\mathrm{n}-1)}{2} + 1$

$\mathrm{m} = \mathrm{n}^2 - \mathrm{n}(\mathrm{n}-1) + 1$

$\mathrm{p} = \frac{\mathrm{n}(\mathrm{n}-1)}{2}$

Given, $\ell + 5 = \mathrm{p} + 2\mathrm{m}$

$\frac{\mathrm{n}^2}{2} + \frac{\mathrm{n}}{2} + 1 + 5 = \frac{\mathrm{n}^2}{2} - \frac{\mathrm{n}}{2} + 2(\mathrm{n}+1)$

$\mathrm{n} + 6 = 2\mathrm{n} + 2$

$\mathrm{n} = 4$

Q90

$f[f(x)] = f(x) \forall x \in S = \{1, 2, 3\}$

I. When range contains 1 element

${}^3C_1 \times 1 = 3$

II. When range contains 2 elements e.g., let $f(1) = 1$ and $f(2) = 2$

For the case of $f(2) = 2$,

Let $f(3) = 1$ OR Let $f(3) = 2$

(i) If $x = 1$,

LHS $= f[f(x)] = f(1)$

RHS $= 1$

In this case also LHS $=$ RHS $\forall x \in S$

(ii) If $x = 2$, LHS $=$ RHS

(iii) If $x = 3$, LHS $=$ RHS

$\therefore {}^3C_2 \times 2 = 6$

Remaining element can be mapped 2 ways.

III. When range contains 3 elements

$f(1) = 1$

$f(2) = 2$

$f(3) = 3$